

# Interface-Induced Trajectory Censoring

A mismatched serving interface distorts both what a benchmark measures  
and what RL training ever sees

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## Abstract

Agent evaluations report a tool-call rate read off the serving stack. That number can be zero while the model is emitting well-formed calls: **the interface censors the trajectory before anything downstream sees it.**

On BFCL v4’s own data, executor and scorer, holding weights, cases, decoding and seeds fixed and changing only the serving adapter configuration, the same model scores **0.00** or **0.96** / **0.19** (`simple_python` / `multi_turn_base`). A  $2 \times 2$  over chat template and parser locates the effect exactly: **both main effects are exactly zero and all of it sits in the interaction** — no component is defective, and repairing one side of the contract buys precisely nothing. On  $\tau$ -bench’s 115 interactive **retail** tasks the same swap moves server-parsed calls from **0 to 636** and tasks reaching any tool execution from **0 to 103**. Our own probe reproduces the funnel across a  $21 \times$  scale range of Qwen2.5-Coder: the server parses **0/100** at every size while well-formed emitted calls rise to **80/100** at 32B ( $\approx 72$  after calibration against a third-party adjudicated gold standard). Under a *matched* envelope, across a comparable span of scale, the silent fraction instead stays at **0–2** — a prediction committed to the repository before the run. Llama-3.1-8B’s 23% rate of calling *the task function itself* as a tool falls to **0** under one `strict: true` flag.

The mismatch reaches inside the training loop, and its consequence is scale-dependent: in verl’s AgentLoop at 7B, **45 of 115 generations carry a complete call; 0 are accepted, 0 execute, 0 return an observation**, while at 1.5B the same zero is over-determined, so we report the two scales separately. Repair restores the mechanism and not the outcome — parsing  $0 \rightarrow 84$ , rescues  $0 \rightarrow 9$ , pass rate  $53 \rightarrow 62$  (n.s.). We release a 98-line preflight check that catches every silent failure reported here. **The tool-call rate is not a property of the model; it is a property of the stack that measures it.**

## 1 Introduction

We set out to train a multi-turn code agent with RL and could not explain our own training curve. Over 150 GRPO steps on Qwen2.5-Coder-1.5B, pass@1 on a held-out EvalPlus split rose 2.6–2.8 points across three independent runs ( $p = 0.064$  /  $0.078$  /  $0.077$  — none significant at  $\alpha = .05$ ; the evidence is the consistent direction across two algorithms and two seeds, not any single test). But **91–94% of newly-passing items passed on the first turn**, and items rescued by turn 2 or later stayed flat at **6–9 out of 540** from step 0 to step 150 (Appendix G). The natural reading is that a 1.5B model cannot learn multi-turn repair. The actual reason is that **the tool was never successfully called**. Not rarely — never. And nothing in the stack said so: the server

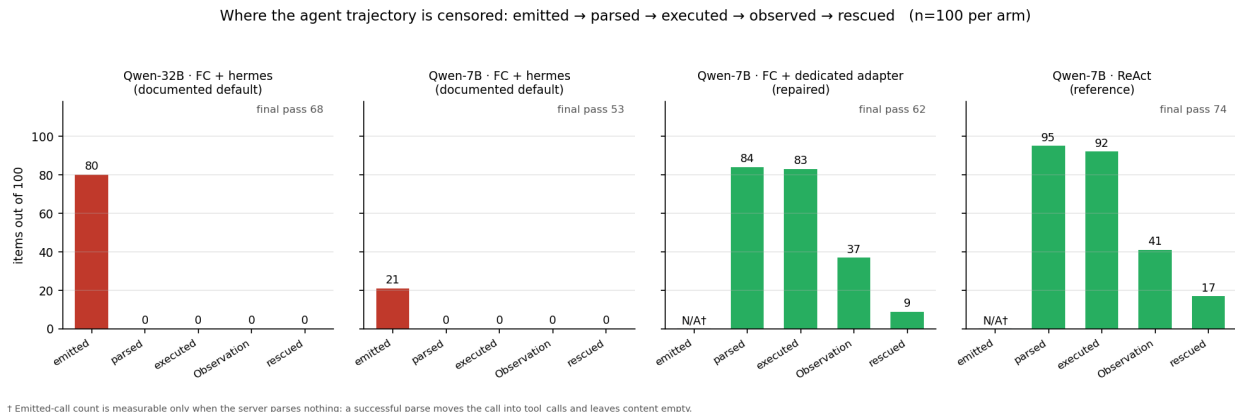


Figure 1: **Where the agent trajectory is censored.** Five layers,  $n=100$  per arm. The two left panels (documented default configuration) show a tall first bar and a cliff to zero: the model emits well-formed calls and nothing downstream happens. Emitted-call counts are measurable only where the server parses nothing — a successful parse moves the call into `tool_calls` and empties `content` — so those cells are N/A, not zero.

returned HTTP 200, `tool_calls` was an empty array, and the training loop recorded a well-formed single-turn trajectory.

An agent evaluation does not measure a model. It measures a composition `model × protocol × serialization × parser × execution stack`, and the failure of most stages is indistinguishable at the output from model incapability: when a parser does not recognise the format a model emits, the serving layer reports zero tool calls — exactly what a model that refuses to use tools would produce. We call this **interface-induced trajectory censoring**.<sup>1</sup> It acts in two opposite directions, and our results contain a clean instance of each: **masking**, where a valid call is not recognised and the action never reaches the environment (§4.3), and **suppression**, where an invalid call is removed from the sampleable support before it becomes an action (§4.5). **The serving interface is therefore simultaneously part of the measurement function and part of the agent’s effective action space** — literally so under RL, where whatever the policy emits must survive serialisation and parsing before it becomes an action that earns a reward. When nothing tool-intended survives that passage, the experience distribution contains no tool-mediated trajectory at all.

**Why this is not a parser bug.** The term invites a reading we refuse at the outset. Replaying vLLM’s own `hermes` extractor line by line over every stored first turn in our archive — 4254 of them, after removing byte-identical duplicate files — we find zero cases in which the server failed to parse a call its own rules would have accepted (§4.3). What fails is the *contract* between three parties configured independently: what the checkpoint was trained to emit, what the chat template tells it to emit, and what the parser is defined to accept. Qwen2.5-Coder emits a well-formed bare call and never the `<tool_call>` envelope, so `hermes` not seeing it is *correct behaviour*; Qwen2.5-Instruct emits the envelope and then breaks the JSON inside it. Censoring can occur at any boundary along `emission → serialization → parse → execution`, and at every one the observable is

<sup>1</sup>Borrowed by analogy: in survival analysis censoring is a threshold on a continuous variable, whereas a parser is a deterministic filter on output *format*. What carries over is the structure of the inference error — the observation is not missing at random but systematically unobservable on one side of a boundary, and the boundary correlates with the quantity being measured.

identical: HTTP 200 with an empty `tool_calls` array. The claim is therefore not that parsers are buggy but that **a tool-call rate is a property of a three-way contract, and is routinely reported as though it were a property of the model alone.**

**What is and is not new.** That Qwen2.5-Coder does not emit `hermes`-format tool calls was reported in vLLM issue #32926 (2026-01-23) with a `<tools>` parser proposal; it was closed as *not planned* and labelled *stale*, so the trap is still live (§2). What this paper adds is measurement, decomposition and consequence. We measure the funnel on *two* standard benchmarks — BFCL v4 and  $\tau$ -bench — where changing only the serving adapter moves BFCL’s reported score from 0.00 to 0.96 / 0.19 and  $\tau$ -bench’s parsed calls from 0 to 636, and a  $2 \times 2$  over template and parser finds both main effects exactly zero with the whole effect in their interaction (§4.1). We separate *model intent* / *emitted format* / *server parse* / *actual execution* / *multi-turn rescue* into five independently measured quantities — reporting only the third is the practice we argue against — quantify what censoring removes across a  $21 \times$  scale range, and pair that ladder with a preregistered counterfactual under a *matched* envelope where the silent fraction stays at 0–2 instead of rising to 80 (§4.3). Four families failing at template, parser, schema and token layers motivate a layered taxonomy on deliberately asymmetric evidence (§4.6). And we follow the distortion into training: at 7B, 45 of 115 generations inside verl’s AgentLoop carry a complete call and none becomes an action, so the sampled experience contains no tool-mediated trajectory to reinforce. We argue that from the sampled rollouts, not the support of the policy — *zero observed samples is not zero probability* — and the comparison arm changes protocol and interface together (§4.9).

## 2 Related Work

**Evaluation validity under nuisance factors.** Reported LLM capability is sensitive to factors that ought to be irrelevant: prompt formatting alone moves benchmark scores by margins comparable to model differences [Sclar et al., 2024], single-prompt evaluation is unreliable [Mizrahi et al., 2024], and benchmark and configuration choices determine which method appears to win [Dehghani et al., 2021]. Those works vary an input the evaluator controls and observe score movement. **We identify a component of the measurement apparatus — the serving stack’s tool-call parser — that can zero out a capability signal entirely, without any error, and whose distortion grows with model scale.**

**Parsing-induced measurement error.** The mechanism has been isolated once before, in another domain: Garware and Zisad [2026] show that in LLM-based security-log classification a strict regular-expression parser reports 0% threat accuracy while a corrected fuzzy parser recovers 76% on *identical* model outputs, with an unaffected severity metric as an internal control. We regard it as the closest antecedent and adopt the same standard of proof: one fixed set of bytes, reparsed under different rules (§4.3). Three things separate the contributions — their gap is a single mechanism in a single domain, which they explicitly decline to generalise, whereas we separate four layers across four families; their gap is a fixed quantity, ours is a function of scale; and their distortion terminates at the reported number, whereas we follow it into an RL rollout and identify which branch of the policy the gradient can no longer reach (§4.8). Outside the literature the claim that tool-call failures originate in serving infrastructure is already practitioner folklore [LiveKit, 2026], qualitatively and without measurements.

**Audits and failure taxonomies of tool use.** A cluster of concurrent 2026 work asks whether reported tool-use scores mean what they appear to mean, and stops one layer above us: Bhat et al. [2026] report an 18.5% evaluator–human misalignment rate across BFCL v4,  $\tau^2$ -bench [Barres

et al., 2025], LiveMCPBench and MCP-Atlas;  $\tau^2$ -Bench-Verified [Amazon AGI, 2025] documents four categories of ground-truth defect in a benchmark used to rank frontier agents; Raj et al. [2026] organise 41 failure modes by interaction edge and by whether the repair belongs to model or harness; Soni [2026] label traces across 1,000 tasks as Tool-Skip, Result-Ignore, Output-Fabrication or Unnecessary-Tool-Use; and Albayaydh et al. [2026] synthesise 27 such papers into six clusters, one being measurement validity. In two cases our result is a precondition for theirs: *Model or Harness?* supplies the vocabulary but no measurement, and *ToolFailBench*’s Tool-Skip label is read as a model behaviour when it is also exactly what an unparsed emission produces — the category is confounded by the interface unless the stack is verified first, which the preflight check of §5 does in seconds. **Every audit above varies something above the API boundary; the distortion we report occurs below it, and is invisible to all of them.**

**Constrained decoding, protocols, stacks.** Tam et al. [2024] report that format restrictions can degrade reasoning; §4.5 is a counter-instance, where `strict: true` eliminates a 23% role-confusion failure and *improves* pass rate, though we also report the CoT control where added reasoning made matters worse. Guided-decoding machinery [Willard and Louf, 2023] underlies both `strict` and vLLM’s `required` mode. ReAct [Yao et al., 2023] is the text protocol we compare against, and BFCL [Patil et al., 2025],  $\tau$ -bench [Yao et al., 2025], ToolLLM [Qin et al., 2024], API-Bank [Li et al., 2023] and Toolformer [Schick et al., 2023] evaluate or train tool use, reporting tool-call rates as parsed by the serving layer. verl [Sheng et al., 2025] provides the RL stack, and importantly its AgentLoop performs **its own** tool-call parsing (`verl/experimental/agent_loop/tool_parser.py`, adapted from vLLM v0.9.1), so a vLLM-side parser plugin does not affect training rollouts — a distinction that cost us one wasted experiment design (§4.9).

**Prior report of the Qwen case.** vLLM issue #32926 [hanXen, 2026], opened 2026-01-23, documents that Qwen2.5-Coder was not trained on tool-call tokens, that it emits `json` fenced blocks or bare JSON absent format instructions, and that a `<tools>` few-shot template plus a dedicated parser reaches 98–100% compliance. It was closed as *not planned* and labelled *stale*. §4.3 reproduces this baseline exactly — zero `<tool_call>` and `<tools>` tags at every scale — and extends it to scale quantification, three further families and training; a related verl issue [ver, 2025] reports the same role-confusion signature we characterise in §4.5. **Our claim is not that the interface can fail silently, which is known; it is that the silent fraction is measurable, that it grows with checkpoint size under the same mismatched interface, and that it propagates into training.** A literature search conducted 2026-09-01 returned nothing that follows the distortion below the API boundary into training; we state this as the scope of our search rather than as a claim of novelty.

### 3 Setup

**Task.** 100 problems from a decontaminated KodCode [kod, 2025] subset (n-gram containment against EvalPlus [Liu et al., 2023]). All 50 full-length arms use the identical 100 items (verified: unique item-set count = 1), so every cross-arm comparison is paired.

**Protocols.** *ReAct* — a text protocol with `Thought: / Action: run_tests / Action Input: / Observation:`, tool name fixed in the template. *Function calling (FC)* — OpenAI-style, `tool_choice: auto`, one tool `run_tests` taking a single `code` string, in three schema variants: terse (bare `{"type": "string"}`), rich (description, counter-example, code sample), strict (`strict: true`, `additionalProperties: false`).

We train with GRPO [Shao et al., 2024] and RLOO [Ahmadian et al., 2024].

**Serving.** vLLM 0.27.1 [Kwon et al., 2023, vll, 2026], per-family parser and chat template as

documented. `temperature=0` is greedy but not bit-wise deterministic under continuous batching; the main results are a single sample per item and Appendix G quantifies sampling variability separately at `temperature=0.6`. `max_tokens = 2048` for both protocols (Appendix A: an earlier asymmetry favoured FC; re-running under the true shared limit changed final pass by  $\leq 1$  item). Serving flags per family and the full training configuration are in Appendix D; every arm’s recorded configuration is in Appendix E.

### 3.1 Intent criterion, and its validation

Reporting only server-parsed calls is the practice under examination, so we need an independent measure of what the model emitted. A single classifier (`analysis/intent.py`) is shared by all text, tables and figures, with three **nested-not-exclusive** tiers: tight — a JSON object naming `run_tests` whose `arguments.code` is a string literal containing real Python, excluding two contaminants we observed (echoing the injected schema; illustrative pseudo-code such as `{"code": test_code}`, an identifier rather than a string); strong  $\supseteq$  tight — trajectory tag `json_named_call` or `xml_tool_call`; and weak, disjoint from both — JSON-shaped fragments not naming the tool, which at 1.5B are almost entirely false positives and are reported separately rather than merged. Every prompt, schema and tier definition is reproduced verbatim in Appendix C.

Three facts about the validation carry into the results; the full two-round procedure, the instrument failure that produced the first round’s  $\kappa = 0.713$ , the adjudication rule and the three successive values of the correction factor are in Appendix F. (i) Against a gold standard built by third-party adjudication of all fifteen non-unanimous items — selected by *any* three-way disagreement, never by disagreement with the classifier — the classifier scores  $\kappa = 0.871$  [0.756, 0.958], and two blind annotators reach inter-annotator  $\kappa = 0.967$  on the 62 items where both read identical bytes. (ii) The **tight** stratum’s precision against that gold standard is  $36/40 = 0.900$ , so §4.3’s headline 80/100 at 32B corrects to  $\approx 72$ ; we report uncorrected counts throughout with this factor stated rather than rescaling silently, and applying it uniformly leaves the trend unchanged (0, 3.6, 18.9, 32.4, 72 against a server-side zero at every size). (iii) The probe stores at most 4000 characters of raw output per item and 10 of the 98 sampled outputs reach that ceiling, so **emitted-call counts are a lower bound** — conservative for our claim, since it can only understate the censoring we report.

### 3.2 Positive control: the pipeline works

Before attributing a zero to a model we verify the instrument. On the same live server, model, tools and prompt, changing only `tool_choice`: under `auto` the server returns `finish_reason = stop` with `tool_calls = 0` while the assistant writes the solution as fenced code; under `required` it returns one parsed call naming `run_tests` with real Python in `arguments`. Server flag, parser and template are therefore all functional. **This establishes that the pipeline can accept a conforming call; it does not locate the zero under auto in the model**, since `required` uses constrained decoding and says nothing about whether the model spontaneously knows the format. The emitted-intent measurement of §4 is what separates them.

### 3.3 Validity gating

An arm is admissible for pass-rate comparison only if it has exactly 100 items, exit code 0, **zero request errors**, and provenance (model / protocol / temperature / `max_tokens` / adapter) matching the intended configuration. Arms with request errors retain a valid error-rate census but are marked N/A for pass rates and *no p-value is computed for them*. **This rule is enforced in code**

(`validate_arms.py`, `final_table.py`), not by convention: an earlier draft computed  $p=0.648$  on such an arm, and that number is retracted (Appendix A).

## 4 Results

Nine measurements, in the order the argument needs them. Each is stated here with its numbers; the tables, the funnel columns behind them and the reproduction records are in Appendix G, and every scope condition is collected in §6.

### 4.1 BFCL v4: the score moves the whole span of the metric, and the effect is pure interaction

BFCL v4 supplies official data, a multi-turn executor and a scorer at pinned Gorilla commit `6ea57973c7a6097fd7c5915698c54c17c5b1b6c8`. Holding weights, cases, decoding, seeds, executor and scorer fixed on 200 cases and changing *only the serving adapter configuration*, BFCL’s own scorer reports **0.00** or **0.96 / 0.19** (`simple_python` / `multi_turn_base`) for the same model. The intervention is a pair — parser *and* its chat template — so we ran the  $2 \times 2$  that separates them.

Table 1: Template  $\times$  parser on BFCL, first request per case.

|                  | documented template | dedicated template |
|------------------|---------------------|--------------------|
| hermes parser    | 0/200               | <b>0/200</b>       |
| dedicated parser | <b>0/200</b>        | 196/200            |

**Both main effects are exactly zero and all of the effect sits in the interaction**, so it cannot be apportioned. Under the documented template the model writes a bare call inside a ````json` fence; under the dedicated template it writes the same payload, byte for byte, inside `<tools>` tags. Each parser behaves correctly for the envelope it is defined to accept, so no component is defective and repairing one side of the contract buys exactly nothing. The funnel behind those scores: the documented arm parses **0 of 200** first requests with zero HTTP errors, while **166** of those same responses carry a call meeting the strict criterion; the repaired arm parses 196, executes 98 of 100 `multi_turn_base` cases and succeeds on **19**, against 0 executed and 0 succeeded for `hermes`, whose zero is therefore *by construction rather than by capability* (Tables 10–20). **A published benchmark number for this model can differ by the entire span of the metric depending on a serving flag the model card does not mention.** We claim only that 0.00 is not a measurement of the model; and since the repaired arm uses the community adapter, not a verified-optimal one, and still fails 4 of 200,  $0 \rightarrow 19$  is a **lower bound** on what the interface was removing.

BFCL’s shipped handler for local Qwen models bypasses the serving parser entirely, so we route BFCL’s own OpenAI handler through `/v1/chat/completions`, adding only `tool_choice: auto` and trace headers; dataset loading, execution and scoring remain BFCL’s code and `bfcl_eval`’s files are byte-identical (Appendix G.4).

### 4.2 $\tau$ -bench: the mechanism survives a simulated user and a stateful environment

BFCL scores a single trajectory against a final state.  $\tau$ -bench [Yao et al., 2025] adds a simulated user and a stateful environment, so a censored call costs not one action but the rest of the conversation.



Across all 115 **retail** tasks, paired across arms with none dropped and changing only the serving adapter: server-parsed calls **0 versus 636**, tool executions **0 versus 636**, and tasks reaching *any* tool execution **0 versus 103**. Every one of the documented arm’s 1389 requests carried **tools**, not one was parsed, and 789 assistant turns carried a call-shaped payload the server discarded (Table 11). **The task-outcome difference does not reach significance and we do not claim it:** 7 versus 10 solved, three discordant pairs all in the repaired direction, exact McNemar  $p=0.25$ , with the documented arm’s solved set a strict subset of the repaired arm’s. Two scope conditions belong with these numbers rather than in a footnote: the user simulator is a locally served Llama-3.1-8B rather than the **gpt-4o** of the original harness, so **the absolute scores are not comparable to the  $\tau$ -bench leaderboard** and only the between-arm difference is claimed; and the arms differ in conversation length by construction, which is why every quantity is per task and not per request.

**The envelope layer is domain-invariant; the payload layer is not.**  $\tau$ -bench also lets us ask whether the *failure layer* travels across domains, because its arguments are short scalars (`{"order_id": "W123"}`) where our own task passes a whole program as a string. Replaying vLLM’s extractor over the documented arm’s 789 emitted-but-unparsed assistant turns puts **789 of 789 (100%) at the envelope layer**, with **0** malformed payloads, **0** recoverable by lenient decoding, and — as everywhere else in this paper — **0 genuine parser losses**. The emissions carry the same signature as the code task: a bare call inside a ````json` fence, never the `<tool_call>` envelope. This matters in two directions. The layer that carries our headline result is **unchanged by the domain**, and the running total of first turns in which the server discarded a call its own extractor would have accepted stays at zero across code and non-code alike. But the *other* layer moves: Qwen2.5-Instruct loses 41% of its unparsed items to malformed payloads on the code task (Table 19), almost always by leaving docstring quotes or raw newlines unescaped inside a `code` string — a failure mode that short scalar arguments cannot produce. **We therefore expect the envelope-layer result to generalise and the payload-layer proportions not to**, and we state that as a prediction rather than a measurement, having tested one non-code domain and not a sample of them.

### 4.3 Censoring against scale, and the same scale under a matched interface

Our own probe, across the Qwen2.5-Coder ladder under `tool_choice: auto` and **hermes** — the documented recommendation for Qwen2.5 —  $n=100$  per size.

Table 2: Qwen2.5-Coder under the documented **hermes** configuration,  $n=100$  per size.

| Size       | Server-parsed | <b>tight</b> | strong ( $\supseteq$ tight) | weak (disjoint) | Final pass |
|------------|---------------|--------------|-----------------------------|-----------------|------------|
| 1.5B       | 0             | 0            | 0                           | 27*             | 31         |
| 3B         | 0             | 4            | 5                           | 28              | 37         |
| 7B         | 0             | 21           | 22                          | 1               | 53         |
| 14B        | 0             | 36           | 42                          | 3               | 53         |
| <b>32B</b> | <b>0</b>      | <b>80</b>    | 100                         | 0               | 68         |

Across a  $21\times$  range the server reports a flat zero while well-formed emitted calls rise to 80/100 ( $\approx 72$  after the 0.900 correction factor of §3.1). **Within this family, the larger the checkpoint, the larger the absolute undercount under the same mismatched interface** — a within-family observation across five checkpoints, not a law relating capability to censoring. Raw-tag

counts reproduce vLLM issue #32926’s baseline exactly: `<tool_call>` and `<tools>` appear zero times at every size. Two offline analyses close the obvious objections (Appendix G.3): re-parsing the *same stored bytes* under four rules excludes “your extractor is simply better than **hermes**”, and replaying vLLM’s extractor line for line to locate each loss shows **genuine parser loss is zero in every arm**, across all 4254 de-duplicated first turns.<sup>2</sup>

**The counterfactual.** The server column is flat zero, so the growth comes entirely from the emitted column — and larger checkpoints emit more well-formed JSON for reasons unrelated to tool calling. Qwen3 supplies a ladder whose template injects tools and requires the `<tool_call>` envelope **hermes** accepts; the prediction (“parsed > 0 at every size”) was committed to the repository before running (p4/PREREGISTRATION.md). Server-parsed counts are 86 / 44 / 70 / 97 at 0.6B–8B, and the silent fraction is **1, 1, 2, 0** against the mismatched ladder’s 0, 4, 21, 36, 80 (Table 12). **Across a comparable span of scale under a matched envelope, the silent fraction does not grow.** This rules out capability growth as a *sufficient* cause; it does not isolate the envelope, since the ladder changes family and envelope together. Two caveats we state rather than bury: the parsed column is **not monotone**, so this shows presence, not a trend; and Qwen3 was served with `enable_thinking=false`, because with thinking on the model exhausts its token budget in a `<think>` block before emitting a call — a confound unrelated to the envelope, recorded as a protocol deviation before the run.

A second family showing the *same* undercount would be stronger, and we did not get one. Of nine families surveyed beforehand, six do not inject OpenAI-style tools at all; of the remaining three, Granite-3.1-8B does not attempt tool calls on this task — 0 under **hermes** and 0 under its own **granite** parser. **That rescue arm is what tells us the arm is uninformative rather than confirmatory**, and we claim nothing from it.

#### 4.4 Within-lineage control, and the ceiling it puts on multi-turn rescue

Changing family cannot test the scale result — Llama parses correctly, Mistral returns 400, DeepSeek never injects the template, so none can exhibit censoring at all. Qwen2.5-**Instruct** can: it shares Coder’s chat template, which is what produces the false-positive template check of §4.6, and differs in whether the checkpoint was trained on tool-call tokens. On the same items, parser, `tool_choice`, temperature and seed, its server-parsed rate runs **1, 11, 63, 77, 88** against Coder’s flat zero (Table 13). **The contrast localises the mismatch to checkpoint-specific training rather than to the shared family and template, or to parameter count alone.** We stop short of naming tool-call-format training as *the* cause: the two differ in their whole post-training mixture, and this varies the bundle, not one element. It also answers the objection that Instruct emits better-shaped calls around worse contents — it solves 13–42% of items on turn 1 and 13–65% by the end, nowhere near the zero that would require.

The multi-turn layer was initially void, because that host lacked `pytest`; we re-ran all five arms with the executor working, changing nothing else and not touching the probe binary, and the parsed column reproduced value for value. **Rescues then rise with the parsed column** — 0, +3, +6, +14, +23 against 1, 11, 63, 77, 88, monotone together. Coder’s first-turn and final passes are identical at every size (31/31, 37/37, 53/53, 53/53, 68/68), which it would be natural to read as multi-turn repair being useless at this scale; the Instruct ladder, on the same items and criterion, shows it is not. We therefore state the repair result as a ceiling rather than an absence: **how much**

<sup>2</sup>The 1.5B row uses the *optional* tool instruction; under the *mandatory* prompt used for training (§4.8) the same checkpoint gives weak = 66 and unparsable = 46. Coercion produces more JSON-shaped debris, not more valid calls.



multi-turn recovers is bounded by how much of the tool channel the interface leaves open.

#### 4.5 A 23% failure that survived three controls and fell to the fourth

Llama-3.1-8B under FC issues a tool call on 97/100 items, but on 23 the call names *the function the task asks it to implement*, with that function’s own parameters as arguments; a re-run restricted to those 23 reproduced 23/23. *This also corrects a measurement of our own*: because the probe did not verify the tool name, these 23 were counted as initiations, so **Llama’s true run\_tests initiation rate is 74/100, not 97/100** (Appendix A). Four single-variable paired controls: a richer schema explicitly warning that the tool is not the task function left it at 22/100 ( $p=1.000$ ); a Thought scaffold before the call raised it to **59** ( $p<0.001$ , *worse*), the model supplying exactly the parameters it had just reasoned about; the official chat template moved nothing (23→22,  $p=0.125$ ); **strict: true** took it to **0** ( $p=0.0001$ ). The first three rejected their alternative explanations and pointed at the model; the fourth found the cause and withdrew that conclusion.

**strict** drives the chain constraint → valid execution (73→97) → task performance (33→46 turn-1, 44→61 final): 23 items that produced no executable code now execute and become re-pairable (Table 14). Paired McNemar decomposes the ReAct-FC gap into **interface repair, +12** ( $p=0.0075$ ) and **protocol, +19** ( $p=0.0019$ ), against +31 for both (Table 15) — **roughly two fifths of what looked like a protocol effect was an interface effect**. With **strict** the two protocols have identical execution mechanics (98 calls, 0 wrong-tool, 97 executed on both sides) yet turn-1 differs 46 vs 61, so what remains is first-draft code quality under the protocol, not tool-use mechanics. The defensible statement is narrower than either “model defect” or “configuration bug”: **the model exhibits role confusion under unconstrained function calling, and the interface constraint determines whether that confusion can be expressed as an environment action**. We cannot show the tendency disappears once constrained, only that it stops reaching the action space.

#### 4.6 Four layers, and what a pre-flight check can see

The two families above fail at different layers with different remedies — Qwen2.5-Coder at the *parser* layer, repaired by a dedicated adapter; Llama at the *schema* layer, repaired by **strict: true** — and neither is caught by the obvious pre-flight test. Two further families extend the taxonomy without supporting a quantitative comparison, and we report them as such. **DeepSeek-Coder** fails at the *template* layer: its chat template never injects tools, which a template check does catch, and we found no remedy. **Mistral-7B-v0.3** fails at the *token* layer, repeating a [TOOL\_CALLS] marker that produces HTTP 400, and has **no admissible 100-item FC run at all** — all four configurations produce request errors (parser only 2; official template 42; parser + strict 3; official template + strict 39), so vLLM’s own recommended configuration *raises* the error rate from 2% to 42% and **strict**, which fixes Llama, does not help (Table 21).

Mistral is the one family whose failure announces itself; the other three return HTTP 200 with **tool\_calls: []**, byte-identical to a model that declined to call the tool. **The claim is not that all failures are silent; it is that the silent ones dominate and are the hardest to attribute**. The natural pre-flight test — does `apply_chat_template(tools=...)` render the schema? — correctly rejects DeepSeek-Coder but *passes* Qwen2.5-Coder, which inherits Hermes scaffolding from Qwen2.5-Instruct while never having been trained on it: **protocol support is a per-checkpoint property and is not inferable from the chat template**.

#### 4.7 Repair loop: how much comes back

On Qwen2.5-Coder-7B with `tool_choice: auto` held fixed and only the adapter changed, parse goes **0**→**84**, items reaching a second turn **0**→**37** and multi-turn rescues **0**→**9**: the mechanism is restored from literal zero (Table 16). The two FC arms have *identical* turn-1 pass, 53 vs 53 — the internal-validity check, since an adapter can only affect what happens after the first turn. That Llama’s turn-1 *does* move (34→46) while Qwen’s does not is the taxonomy making a prediction: Qwen’s failure is downstream of generation, so the first turn is produced and then discarded; Llama’s is inside generation, so the first turn is itself destroyed and repairing the constraint restores it. **Different failure layers leave different downstream signatures, and the direction of the turn-1 effect identifies the layer.**

The pass-rate gain is *not* significant (53→62,  $p=0.093$ ). A pre-planned replication on a random 300-item sample reproduces the direction (160/300→178/300, 43 discordant against 25,  $p=0.0385$ ) — nominally below 0.05, but this paper runs roughly a dozen McNemar tests and the Bonferroni threshold is  $\approx 0.0042$ , so **the repair gain still does not survive correction and we rest no claim on it.** The protocol gap on the same 300 items (178/300→219/300,  $p=3.1\times 10^{-6}$ ) clears it by three orders of magnitude. Restricting to items both arms parsed, the residual protocol gap holds on Llama ( $p=0.0021$ ) and is not detected on Qwen: +8.4 pp, 95% CI  $[-0.5, +17.4]$ ,  $p=0.118$ . **We do not read  $p > 0.05$  as evidence of no effect** — that interval admits a residual larger than the effect we detect on Llama — so the residual is established on one family and undetermined on the other (Appendix G.8).

#### 4.8 Zero tool execution in RL rollouts, and where it comes from at two scales

With model, data, hyper-parameters, seed and prompt strength matched, the FC arm’s `num_turns` is pinned at its minimum of **2.000**, tool time is **0.00 s** and a dedicated counter reads **0.000** calls per rollout, against 5.883 / 12.21 s / 2.052 for ReAct (Table 17). Instrumentation sits at the two agent-side call sites and *not* at the sandbox entry point, because the reward function evaluates final code through the same function and would merge reward evaluation into the tool-call count. **Wording bound.** What is measured is **parser-accepted and executed calls = 0**, *not* “the model never attempted to call”: FC rollouts were not saved.

At 1.5B that zero is **over-determined**. A probe replication under the identical mandatory prompt gives 0 well-formed calls, 54 direct answers, 46 unparsable outputs and 66 items containing JSON fragments but *not one naming `run_tests`* — a capability failure, not a parser one. So we ran the probe again inside verl’s own AgentLoop at 7B, under the documented configuration, logging every generation, the parser’s verdict, executions and observations.

Table 3: Rollout-path probe inside verl, Qwen2.5-Coder-7B, documented **hermes** configuration, 115 generations.

|                                                           | count     |
|-----------------------------------------------------------|-----------|
| bare JSON <code>run_tests</code> call, valid and complete | <b>45</b> |
| names <code>run_tests</code> but payload malformed        | 12        |
| no call attempted                                         | 58        |
| accepted by verl’s <b>hermes</b> parser                   | <b>0</b>  |
| tool executions                                           | <b>0</b>  |
| observations returned                                     | <b>0</b>  |

**Forty-five of 115 generations carry a call that would execute if it were wrapped in the**

**envelope, and none reaches the environment.** This is the serving-side mechanism of §4.3 observed *inside the training loop*, which is where the RL claim needs it: at 7B the absence of tool-mediated trajectories is attributable to the interface, not the policy. It does not retroactively change the 1.5B run, whose zero remains over-determined, and we report the two scales separately. This probe’s run terminated with a **CUDA OOM** during the actor update following rollout collection, so no gradient step completed; the rollout observations all precede that point.

Valid calls are zero **from step 1** and the reward admits direct answers (FC **critic/rewards/mean** rose 0.233→0.281), so the tool branch has a zero-sample return while direct answering is continuously reinforced. The model cannot discover the tool path by exploration, because every attempt is discarded at the protocol layer and no positively-rewarded tool trajectory ever exists. **The tool-using branch receives no direct on-policy gradient signal under the observed rollout distribution, and is effectively inaccessible to policy-gradient learning in this cold-start regime.** At 10 and 3 steps this is a *mechanism* result, not an outcome comparison.

#### 4.9 Why no repaired-FC arm exists yet, and what a working channel did not buy

The natural control is *repaired* FC rather than ReAct, and we did not have it at the time of writing. Four obstacles were met and three dissolved on inspection (Appendix D.5). A vLLM parser plugin does not reach the training loop at all, because verl’s AgentLoop parses for itself. A parser registered inside *verl*’s registry does reach it and lifts recoverable calls from 0/100 to **52/100** at 1.5B — but **zero of those 52 name run\_tests**, every one targeting the function the task asks the model to write, which name normalisation cannot repair because no call carries code to normalise. A role-disambiguation few-shot aimed at that confusion made the model *more fluent at emitting the wrong call* (52→64 recoverable, still 0 naming the tool) and slightly worse at the task (15→13).

**Correcting our own prescription.** We previously named the missing capability as guided decoding. That was the wrong diagnosis: guided decoding *forces* a format, whereas the repair argued for everywhere else here is the opposite — accept what the model already emits. **What actually blocks the arm is scale:** at 1.5B a repaired parser would have nothing to accept, and the checkpoint that does emit well-formed calls is 7B, whose full-parameter RL exhausted memory on our single 80 GB card. The control is a 7B experiment under parameter-efficient tuning, not a stack-replacement project (§6).

Accordingly **ReAct serves as a positive-control interaction channel, not a parser-repair control**, and it is the only configuration delivering tool feedback at this scale. Run for 75 GRPO steps with everything else matched to the historical runs, it executed **23,676 tool calls** and left rescues by turn  $\geq 2$  at **8, 6, 6, 8, 6** over steps 0–60 — net change zero, inside the range of the zero-execution runs. **The interface determines whether tool-mediated trajectories exist at all, and their absence is sufficient to explain why no multi-turn signal was available — but their presence is not sufficient to produce multi-turn learning at this scale and budget.** Because this arm changes protocol and interface together it cannot separate “a working channel is insufficient” from “ReAct in particular is insufficient”; it is 75 steps on one seed at 1.5B against 150-step baselines, and the step-75 evaluation was lost when the checkpoint write exhausted the disk after training completed.

## 5 Discussion

**Report the composition, not the model.** Any claim of the form “model  $M$  does not use tools” is, as measured, a claim about  $M \times \text{protocol} \times \text{serialization} \times \text{parser} \times \text{stack}$ . Agent evaluations should report at minimum (i) server-parsed calls, (ii) emitted-but-unparsed calls under a stated criterion, (iii) request errors, and (iv) the exact serving configuration. Reporting only (i) is the practice this paper argues against, and (ii) is where the  $21\times$  scale trend lives.

**Pre-flight, don’t post-hoc.** A missing `--enable-auto-tool-choice` is the only failure that announces itself, and template inspection alone is insufficient (§4.6). We therefore treat a `tools-`bearing request returning HTTP 200 — *and* a server-parsed call under `tool_choice: required` — as a required pre-flight for any FC experiment, shipped as a 98-line script (`preflight_toolcall.py`) that issues one canonical request, asserts `tool_calls` is non-empty with the expected `name` and parseable `arguments`, then repeats under `required` as a positive control. It runs in seconds and would have caught every silent failure in this paper.

**The guidance for this flag lives in the wrong place.** We surveyed the HuggingFace model cards of nine families for any mention of `--tool-call-parser`. Of the eight reachable cards, none mentions it, while six recommend serving the model with vLLM. The publisher tells you which server to use, the server asks you which parser to use, and neither treats the pairing as its responsibility. We do *not* read this as evidence that mismatch is common in deployment — we surveyed documents, not deployments, and vLLM ships dedicated parsers for some forty families, so following *its* table generally works. The claim is narrower: **the one piece of configuration that can silently zero a capability measurement is documented by neither party at the point of use.**

**Interface repair precedes protocol comparison.** On Llama the ReAct-vs-FC gap was 31 points before the `strict` fix and 19 after, both components separately significant. On Qwen, conditioning on parsed items, we do not detect a residual ( $p=0.118$ ) — but that interval is wide enough to contain effects as large as the unconditioned gap, so it is evidence of *insufficient resolution*, not equivalence. **Comparisons of agent protocols that do not first exhaust the serving configuration matrix are not measuring protocols.**

**A cold-start hazard for agentic RL.** §4.8’s mechanism generalises beyond this stack: whenever (a) tool trajectories are censored at rate  $\sim 1$  and (b) the reward admits a tool-free path, the tool-using branch has no sampled return and cannot be reinforced. Prompt-level coercion does not help, and we observed it actively hurting (final pass 31→15). Practitioners should verify a **non-zero tool-execution count in the first training step** before interpreting any multi-turn result.

## 6 Limitations

Seventeen limitations, in the order a sceptical reader should apply them, are stated in full in Appendix H; the six that most constrain what may be concluded are here.

**The scale result is within one family.** The  $0 \rightarrow 4 \rightarrow 21 \rightarrow 36 \rightarrow 80$  ladder is Qwen2.5-Coder alone, against one mismatched interface, on one task (item 3). We claim a within-family monotone relation between checkpoint size and absolute undercount; we do not claim censoring is generally scale-increasing, and we do not equate parameter count with capability. A second *family*’s ladder is the test that would promote or refute it.

**There is no clean repaired-FC training control** (item 8). The training evidence compares FC against ReAct, changing protocol and interface together. We previously attributed the arm’s

absence to verl lacking guided decoding; **that was wrong** — the registry is extensible and we have since installed code in it — and what blocks it is scale (§4.9). We therefore claim that interface censoring can be observed directly inside rollout collection and eliminates tool-mediated samples; we do **not** claim to have causally proven that it prevents RL from learning multi-turn repair.

**Observed zero is not zero probability, and zero executions is not zero attempts** (items 10–11). Our instrumentation records the sampled experience distribution, not the support of the policy, and the FC training arm did not save raw rollout text, so “never attempted” cannot be separated from “attempted and discarded”. Retaining rollout text is a change we would make before any further training.

**Conditioning on a successful parse is post-treatment conditioning** (item 9). Whether an item parses is determined by the interface under test, so the both-parsed subset is not a random subsample. On Qwen the residual is +8.4 pp, 95% CI  $[-0.5, +17.4]$ ,  $p=0.118$ : **we do not detect a residual difference, which is not the same as establishing there is none.**

**Multiple comparisons** (item 6). Roughly a dozen McNemar tests; the Bonferroni threshold is  $\approx 0.0042$ . Surviving it: the `strict: true` intervention ( $p=0.0001$ ) and the protocol gap on 300 random items ( $p=3.1 \times 10^{-6}$ ). **Not surviving it:** the repair-loop pass gain ( $p=0.093$  at  $n=100$ ,  $p=0.0385$  at  $n=300$ ), the Llama protocol contrast at  $n=300$  ( $p=0.0352$ ), the  $\tau$ -bench outcome difference ( $p=0.25$ ), and the residual-gap estimates ( $p=0.118, 0.125$ ). We rest no claim on the latter group.

**The training verifier has two exploitable seams, which we audited rather than assumed away** (item 14). The summary parser takes the first `(\d+)` `passed` match over merged `stdout/stderr`, and `skipped/xfail` are not in the denominator; both are reachable in principle. Across the five training logs that retain generated code there are **zero** occurrences of a printed ```N passed''`, of `pytest.skip` or of `sys.exit`, and `critic/rewards/mean` oscillates in 0.46–0.73 without the step change a discovered exploit produces. **The audit’s own limit is that full rollout text was not retained**, so it covers only the fragments those logs carry.

Also constraining, and stated in full in the appendix: one task family and one tool (item 1); a non-random 100-item set, whose 7B rung the random 300-item replication moves from 21 to 30 (item 2); a deliberately asymmetric taxonomy in which Mistral has no admissible FC comparison at all (item 4); single-sample headline arms (item 5); scale stopping at 32B (item 7); sufficiency tested only through ReAct (item 12); a stratified validation sample with a 4000-character output cap that makes emitted-call counts a lower bound (item 13); dependence on vLLM 0.27.1, verl 0.9.0 and publisher-mutable chat templates (items 15–16); and the instrumentation errata we disclose rather than absorb — seven arms recording `max_tokens` they did not run at, Llama’s initiation rate corrected from 97 to 74, and one previously reported  $p=0.648$  withdrawn (item 17, Appendix A).

## What would change our conclusions

We name these so the claims are falsifiable rather than merely hedged. *A broken-FC versus repaired-FC RL comparison*, everything else fixed — a parser registered in verl’s own registry, two arms differing only in that registration, run at **7B** under parameter-efficient tuning because 1.5B emits nothing for a repaired parser to accept and 7B full-parameter RL does not fit our card. **If repaired-FC recovers multi-turn learning, our sufficiency claim in §4.9 is wrong**; if it does not, the “bottleneck is not singular” reading strengthens from one arm to two, the second being single-variable. This is the experiment most likely to overturn something we have written, which is why we name it precisely. *A second family’s scale ladder*, with a working test executor: if the undercount does not grow with checkpoint size elsewhere, §4.3’s claim is a Qwen-plus-hermes fact rather than a scaling phenomenon. *The same ladder on a non-code task*. The five Qwen2.5-Coder checkpoints



run against  $\tau$ -bench **retail** under the documented configuration — models and interface fixed, only the domain varied. If the emitted column still rises with size while the parsed column stays at zero, the scale result is a property of the interface; if it does not, it is a property of code-shaped payloads. This needs a rescue arm at the small sizes for the reason Granite did: a checkpoint that simply never attempts a call produces an uninformative zero, not a censored one. *The same five-layer measurement on further benchmarks.* Done for BFCL v4 and  $\tau$ -bench **retail**, so the concern is not specific to our harness and now covers one interactive suite; suites with much larger tool schemas, or whose scoring depends on argument-level matching, remain untested.

## 7 Conclusion

Tool-call interfaces censor agent trajectories, silently and per checkpoint; within Qwen2.5-Coder the absolute undercount rises monotonically with checkpoint size, which we report as a within-family observation rather than a general law. In reinforcement learning the same mismatch leaves the sampled experience distribution with no tool-mediated trajectories — 45 of 115 generations inside the training loop carry a complete call and none becomes an action — so the tool-using branch receives no direct on-policy gradient signal under the observed rollout distribution. Repairing the interface restores the mechanism but recovers only part of the outcome gap.

Two controls fix what the effect is a property of. A  $2 \times 2$  over chat template and parser on a public benchmark finds both main effects exactly zero, with the entire 0.00-to-0.96 swing in their interaction; and a ladder under a *matched* interface holds the silent fraction at 0–2 across a comparable span of scale, where the mismatched interface takes it to 80. **What the serving layer removes is not a property of the model, nor of the parser, but of the contract between them.**

The most uncomfortable finding is methodological. We ran three single-variable paired controls on a 23% failure and concluded it was a model deficiency; a fourth, taken from a sentence in the serving documentation, reduced it to zero. Our own human validation reproduced the same mechanism: twelve of ninety-eight annotations missed a tool call because it sat after a fenced code block, and the annotator — like the parser — concluded from the same bytes that no call had been made. Reporting server-parsed tool-call rates as model capability is not a safe default.

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## A Data Errata

Every number in this paper traces to a trajectory file under `runs/final/`. This appendix lists the files whose recorded provenance is wrong, the arms that are inadmissible for pass-rate comparison, and what was verified correct. It is reproduced from `runs/final/ERRATA.md` in the released repository.

### A.1 Seven ReAct arms record the wrong max\_tokens

The following files record `max_tokens=2048` while the generation limit in force was **1024**. Cause: `gen()` in the probe carried a default of `max_tokens=1024` while `gen_fc()` used the global `MAX_TOKENS=2048`, and the provenance written to disk recorded `MAX_TOKENS` for both. Fixed 2026-08-31.

|                                              |                             |
|----------------------------------------------|-----------------------------|
| <code>traj_v3_Llama8B_react.jsonl</code>     | recorded 2048 / actual 1024 |
| <code>traj_v3_Qwen7B_react.jsonl</code>      | recorded 2048 / actual 1024 |
| <code>traj_v3_Mistral7B_react.jsonl</code>   | recorded 2048 / actual 1024 |
| <code>traj_v11_DS1.3b_react.jsonl</code>     | recorded 2048 / actual 1024 |
| <code>traj_v11_DS6.7b_react.jsonl</code>     | recorded 2048 / actual 1024 |
| <code>traj_v11_Llama32_1B_react.jsonl</code> | recorded 2048 / actual 1024 |
| <code>traj_v11_Llama32_3B_react.jsonl</code> | recorded 2048 / actual 1024 |

**Direction of the bias.** The contemporaneous FC arms were genuinely at 2048, so in every affected ReAct-vs-FC comparison **the function-calling side had the larger generation budget**. ReAct nonetheless won those comparisons, so the reported protocol gaps are *conservative*; correcting the asymmetry can only widen them. Re-running the three main ReAct arms at a true 2048 changed final pass by at most one item ( $80 \rightarrow 80$ ,  $74 \rightarrow 74$ ,  $32 \rightarrow 33$ ), which is why the correction is small in practice.

**Replacement data.** The four v11 arms are superseded by `traj_v13*_react.jsonl`; the three v3 arms by `traj_v14*_react.jsonl`. All tables in this paper cite the v13/v14 versions; the v3/v11 ReAct arms are retained only as a record.

### A.2 Llama-3.1-8B initiation rate required manual correction

`traj_v3_Llama8B_fc.jsonl` records an initiation rate of 97/100, but 23 of those calls target **the task function rather than run\_tests**. The probe did not verify tool names when that batch was collected, so `has_action` did not distinguish them. **The corrected run\_tests initiation rate is 74/100.**

A re-run restricted to exactly those 23 items under matched configuration (`traj_v8_Llama8B_recheck.jsonl`, seed fixed) reproduced 23/23 as wrong-tool calls, with names such as `can_form_word`, `check_password_strength` and `floyd_warshall`.

Of the same batch, `traj_v3_Qwen7B_fc_nosuffix.jsonl` has an initiation rate of 0 and needs no correction; `traj_v3_Mistral7B_fc.jsonl` has 2, and those two were not individually re-checked for tool name.

### A.3 Arms inadmissible for pass-rate comparison

|                                                         |                         |                     |
|---------------------------------------------------------|-------------------------|---------------------|
| <code>traj_v3_Mistral7B_fc.jsonl</code>                 | <code>n_err = 2</code>  | <code>(rc=2)</code> |
| <code>traj_v5_Mistral7B_fc_official.jsonl</code>        | <code>n_err = 42</code> | <code>(rc=2)</code> |
| <code>traj_v9_Mistral7B_fc_strict.jsonl</code>          | <code>n_err = 3</code>  | <code>(rc=2)</code> |
| <code>traj_v9_Mistral7B_fc_official_strict.jsonl</code> | <code>n_err = 39</code> | <code>(rc=2)</code> |
| <code>traj_v13_Llama8B_fcstrict_t06_s3.jsonl</code>     | <code>n_err = 1</code>  | <code>(rc=2)</code> |

For these arms the **error-rate census remains valid** — that census is itself a result — but pass rates are not comparable because data are missing, and no *p*-value is computed for them. An earlier draft computed  $p = 0.648$  on one such arm; that number is retracted.

## A.4 Provenance schema drift

Fields were added over the course of the work, giving four schema generations: early arms lack `seed`, `temperature` or `fc_schema`. The fields load-bearing for every claim (`model`, `protocol`, `adapter`, `max_tokens`, `clean_index`) are present in all generations. A `script_sha256` field was never successfully added — the patch was interrupted twice — so script identity across a run is instead guaranteed externally, by hashing the probe before launch (`v13_pinned_hashes.txt`) and re-checking after.

## A.5 Verified correct

- All 50 full-length arms use the identical 100-item set (`clean[:100]`; unique item-set count = 1), so every cross-arm comparison in this paper is exactly paired.
- Every FC arm ran at a true `max_tokens` of 2048.
- Sampling in the variance experiment is genuine: at `temperature` = 0.6, two seeds differ in first-turn code on 95/100 items and flip outcomes on 16/100.

## A.6 Third-party component versions

The Qwen2.5-Coder `<tools>` parser is taken from `hanXen/vllm-qwen2.5-coder-tool-parser` (Apache 2.0), downloaded 2026-08-31, at the then-current main:

```
commit 1b921501f30cbfe347dccbd7de3c82a1d55131 (1b92150, 2026-04-29)
    "fix: buffer partial <tools> prefix in streaming to prevent tag leak"
```

```
SHA256 (first 20):
c16bb1f88936a2d96c7c  qwen2_5_coder_tool_parser.py          (unmodified)
736bd175adbf90942c1d  tool_chat_template_qwen2_5_coder.jinja (MODIFIED)
a95b2a9b91e65b3d452f  tool_chat_template_qwen2_5_coder.jinja.orig (original)
```

The template was modified because its few-shot examples invoke `get_weather`, a tool absent from our schema. Left unchanged it would induce calls to a non-existent tool, simultaneously inflating the initiation count and contaminating the empty-argument statistic. The original is retained alongside.

## B Reproducibility

```
# analysis and figures (no GPU)
pip install -r requirements.txt
python analysis/intent.py          # the single intent criterion, all four counts
python runs/final/final_table.py  # main / variance / error-rate tables (N/A enforced)
python figures/make_figs.py

# probe (needs a vLLM server)
python probe_react_full.py --model <path> --port 8000 --n 100 \
    --protocol {react,fc} --strength {optional,mandatory} \
    --fc-schema {terse,rich,strict} --parser-adapter cross_family \
    --temperature 0.0 --seed 0 --out traj.jsonl
python validate_arms.py           # per-arm admissibility: lines / rc / n_err / provenance / script ha
```

Model snapshots must be pinned: `tokenizer_config.json` inheritance is the mechanism behind §4.6’s false-positive capability check, so a family’s behaviour can change with a repository revision. Every model path, its HF revision, and the serving flags used are recorded per arm in the trajectory provenance and in `runs/final/by_config/README.md`.

Trajectories for all 50 full-length arms, the training logs, the errata, and the per-configuration index are in `runs/final/`. Third-party parser: hanXen (1b92150, Apache 2.0), few-shot examples rewritten from `get_weather` to `run_tests`; original retained with both hashes recorded.

## C Prompts and tool schemas, verbatim

All prompts are reproduced exactly as sent, in the original Chinese. We do not translate them: the protocol comparison is between two prompts, and a translation is a third prompt. An English gloss follows each block for readers who do not read Chinese, but the gloss was never sent to any model.

### C.1 ReAct

Two strengths. `optional` permits a direct answer; `mandatory` requires an `Action` first. The main table uses `optional` on both protocols so that neither side is coerced.

REACT\_OPTIONAL

-----

你是一个 Python 编程助手，可以按以下格式工作：

Thought: 你的思考

Action: `run_tests`

Action Input: ```python

<完整代码>

```

Observation: (由系统填写测试结果)

Thought: 我已确认代码正确

Final Answer: ```python

<最终代码>

```

你可以用 `Action` 提交代码验证，也可以直接给 `Final Answer`。

*Gloss:* “You are a Python coding assistant and may work in the following format: Thought / Action: `run_tests` / Action Input: (fenced code) / Observation (filled in by the system) / ... / Final Answer: (fenced code). You may submit code with Action for verification, or give a Final Answer directly.”

The mandatory variant is identical except that the closing line becomes **必须先用 Action 提交代码验证，再给 Final Answer**。（“You must first submit code for verification with Action, then give a Final Answer”), and the loop is marked as repeatable.

### C.2 Function calling

FC\_OPTIONAL

-----

你是一个 Python 编程助手。请根据用户的描述实现所要求的函数。

你可以调用 `run_tests` 工具把代码交给测试运行，工具会返回通过情况与报错，

你可以据此修改代码再次提交。

FC\_MANDATORY

你是一个 Python 编程助手。请根据用户的描述实现所要求的函数。

**\*\*你必须先调用 `run_tests` 工具验证你的代码\*\***，不要直接给出答案。  
工具会返回测试通过情况与报错；若有失败，请修改代码并再次调用 `run_tests`。

*Gloss:* optional — “You may call the `run_tests` tool to submit code to the test runner; it returns pass/fail and errors, and you may revise and resubmit.” mandatory — “**You must first call the `run_tests` tool to verify your code**; do not answer directly.”

**Strength is matched across protocols in every comparison we report.** The training comparison in §4.8 uses mandatory on both sides, so “FC executed zero tool calls” cannot be attributed to a weaker instruction.

### C.3 Tool schemas: terse, rich, strict

`terse` is the schema a reader arrives at by following the API documentation. It is the schema under which Llama’s 23% wrong-tool rate appears.

```
terse
-----
{"type": "function", "function": {
  "name": "run_tests",
  "description": "把你写的 Python 代码交给测试运行，返回通过情况与报错。",
  "parameters": {"type": "object",
    "properties": {"code": {"type": "string"}},
    "required": ["code"]}}
```

`rich` is the §4.5 control. Its purpose is to test whether the wrong-tool rate is simply an under-specified schema, so it is written to be maximally explicit: the description states that the tool is *not* the function the user asked for, and the parameter description carries a worked example.

```
rich
-----
{"type": "function", "function": {
  "name": "run_tests",
  "description": "把你写的 Python 代码提交给测试运行器执行，返回测试通过情况与报错。"
    "注意：这个工具不是你要实现的那个函数，不要把题目函数的参数传进来；"
    "唯一的参数 code 是你写的完整源代码文本。",
  "parameters": {"type": "object",
    "properties": {"code": {
      "type": "string",
      "description": "完整的 Python 源代码文本，包含所需的 import 和完整的"
        "函数定义。例如：\n\"def add(a, b):\n    return a + b\n\""},
      "required": ["code"]}}
```

*Gloss of the added text:* “Note: this tool is not the function you are being asked to implement. Do not pass the task function’s parameters into it. Its only parameter, `code`, is the complete source text you wrote.”

**Under this explicit warning 22/100 items still pass the task function’s parameters**, and they are the same parameter names that failed under `terse` (§4.5,  $p=1.000$ ).

`strict` is `terse` plus `"strict": true` on the function object, with `VLLM_ENFORCE_STRICT_TOOL_CALLING=true` (the default). This is the single change that takes the wrong-tool rate to zero.



## C.4 The Thought-scaffold control (§4.5)

To test whether ReAct’s advantage is merely an extra reasoning step, the FC system prompt was given a Thought requirement and nothing else changed:

“First write out your reasoning in a Thought: what the problem asks for, what the edge cases are, what algorithm you intend to use. Once you have thought it through, call the `run_tests` tool...”

The wrong-tool rate rose from 23 to 59 ( $p < 0.001$ ). We report this as a single wording, not as a claim about reasoning scaffolds in general.

## C.5 The role-disambiguation few-shot (§4.8)

Used at training scale (1.5B) after the wrong-tool failure was found to be semantic rather than syntactic. It shows the wrong call and the right call side by side.

你是一个 `Python` 编程助手。请根据用户的描述实现所要求的函数。

**\*\*你必须先调用 `run_tests` 工具验证你的代码\*\***，不要直接给出答案。

注意一个常见错误：**\*\*用户描述里要你实现的那个函数，不是可调用的工具。\*\***  
你唯一可以调用的工具是 `run_tests`，它只有一个参数 `code`，内容是你写的完整源代码。

错误示范（用户要求实现 `join_integers`）：  
`{"name": "join_integers", "arguments": {"int_array": [1, 2, 3]}}`  
—— 这是在调用你自己该编写的函数，`run_tests` 收不到任何代码。

正确示范：  
`{"name": "run_tests", "arguments": {"code": "def join_integers(int_array):\n return ','.join(map(str, int_array))"}}`  
—— 函数名恒为 `run_tests`，你写的整段代码作为字符串放进 `code`。

工具会返回测试通过情况与报错；若有失败，请修改代码并再次调用 `run_tests`。

**It made things worse:** recoverable calls rose  $52 \rightarrow 64$  while calls naming `run_tests` stayed at 0 and final pass fell  $15 \rightarrow 13$ . The model became more fluent at emitting the wrong call.

## C.6 The intent criterion

Every emitted-call count in this paper comes from one function, applied identically to every arm. There is no per-arm tuning.

```
TIGHT = re.compile(
    r'"name"\s*:\s*"run_tests"\.{0,200}?"arguments"\s*:\s*\{'
    r'\.{0,80}?"code"\s*:\s*"(.{0,4000}?)"\s*\}', re.S)

REAL = re.compile(r'\n\def |class |return |import |lambda ')

def is_tight_call(raw):
    m = TIGHT.search(raw or "")
    return bool(m and REAL.search(m.group(1)))
```

TIGHT requires the name, an `arguments` object, and a `code` key whose value is a *string literal*; REAL then requires that literal to contain actual Python. Together they exclude schema

echo (parameters/description present, arguments absent), identifiers passed instead of strings ("code": test\_code), and placeholders ("<your code here>").

The **strong** tier drops the payload requirement (name present, or an XML tool tag); **weak** counts JSON structure with no `run_tests` name and is disjoint from both. Tiers are nested, not mutually exclusive, and the paper states which tier each number uses. Human validation of this criterion is in §3.1; two independent blind annotators give inter-annotator  $\kappa = 0.967$  on items where both read identical bytes.

## D Serving and training configuration

### D.1 Per-family serving flags

These are the configurations under test, not our inventions: each is what the vLLM documentation prescribes for that family. The rightmost column is what happens when the flag is absent — the property that motivates the paper.

Table 4: Serving configuration per family (vLLM 0.27.1) and the symptom of omission.

| Family              | Flags                                                                                                                                                                                                                                                               | If omitted                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| all                 | <code>--enable-auto-tool-choice</code>                                                                                                                                                                                                                              | HTTP 400 — <b>the only loud failure</b>                                                            |
| Qwen2.5 (In-struct) | <code>--tool-call-parser hermes</code>                                                                                                                                                                                                                              | silent empty <code>tool_calls</code>                                                               |
| Qwen2.5-Coder       | <i>no working stock configuration</i> ; we install the community <code>&lt;tools&gt;</code> parser plus its chat template                                                                                                                                           | silent, at every scale                                                                             |
| Llama 3.1 / 3.2     | <code>--tool-call-parser llama3_json</code><br><code>--chat-template tool_chat_template_llama3.1_json.jinja</code><br>plus <code>"strict": true</code> in the schema                                                                                                | malformed arguments; without <b>strict</b> the wrong-tool rate is 23%                              |
| Mistral (HF format) | <code>--tokenizer-mode hf --config-format hf</code><br><code>--load-format hf</code><br><code>--tool-call-parser mistral</code><br><code>--chat-template tool_chat_template_mistral_parallel.jinja</code><br><code>tool_call_id</code> must be exactly 9 characters | HTTP 400; the officially recommended parallel template <i>raises</i> the error rate from 2% to 42% |
| DeepSeek-Coder      | none available: the chat template never injects <code>tools</code>                                                                                                                                                                                                  | n/a                                                                                                |

**Only the first row fails loudly.** Every other omission returns HTTP 200 with an empty `tool_calls` array, which is byte-identical to a model that declined to call the tool. This is what `preflight_toolcall.py` (98 lines, released with this paper) exists to catch: it issues one canonical request, asserts `tool_calls` is non-empty with the right name and parseable arguments, then repeats under `tool_choice: required` as a positive control. It would have caught every silent failure reported here.

### D.2 Training configuration

Shared by all training arms, including the three historical runs of §1 and the arms of §4.8:

Table 5: Training configuration. All arms share every value below; arms differ only in the protocol and, for the historical runs, the algorithm and seed.

| Setting               | Value                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------|
| Model                 | Qwen2.5-Coder-1.5B-Instruct                                                               |
| Framework             | verl 0.9.0 (vLLM 0.27.1 rollout backend)                                                  |
| Algorithm             | GRPO ( $\times 2$ seeds) and RLOO ( $\times 1$ seed)                                      |
| Trajectories per step | <code>train_batch_size</code> $16 \times \text{rollout.n}$ $8 = 128$                      |
| Steps                 | 150 (historical runs); 10 / 3 / 75 for the arms of §4.8                                   |
| Training data         | KodCode, decontaminated pool of 7669 items                                                |
| Held-out evaluation   | EvalPlus: HumanEval+ 319, MBPP+ 677                                                       |
| Denominators          | 540 (multi channel), 454 (repair channel), after removing 2 known-defective items         |
| Agent loop            | <code>tool_agent</code> (FC) or <code>react_agent</code> (ReAct)                          |
| Tool                  | <code>run_tests</code> , single <code>code</code> argument, sandboxed <code>pytest</code> |
| System prompt         | FC_MANDATORY / ReAct mandatory — matched strength (Appendix C)                            |
| Hardware              | single A800 80 GB                                                                         |

### D.3 Tool-call instrumentation

verl 0.9.0’s step metrics contain **no** tool-call count. The metric named `timing_s/agent_loop/tool_calls` is *elapsed seconds*; reading it as a count is a hard error and we flag it because the name invites exactly that.

We therefore instrumented the two agent-side call sites directly, each writing one labelled line per call:

- `CodeTool.execute` — the function-calling path
- `ReActAgentLoop._run_tests` — the ReAct path

**The counter must not be placed in `sandbox.run_tests`.** The reward function evaluates the final program through that same function, so a counter there conflates reward evaluation with agent tool use. We hit this and moved the counter; the sandbox’s own counter was renamed `sandbox_exec_count` so the two can never be confused again.

Result: the FC arm’s counter file has **0 lines**; the ReAct arm’s has **788**, all carrying the `react` label, so the provenance is uncontaminated. Three mutually independent signals agree — `num_turns` pinned at its minimum of 2, tool time exactly 0.00s, and a dedicated counter at 0.

### D.4 Why no repaired-FC training arm exists

For completeness, the three layers that block the control experiment a reader will ask for (§4.8, and Limitations item 8):

1. `multi_turn.format=hermes` selects **verl’s own** `ToolParser` registry, which is independent of vLLM’s — so installing a vLLM parser plugin has no effect on the training path.
2. A verl-side parser we wrote does recover calls, but **none of them are the tool**: 36/52 target the task function, 16/52 carry executable code elsewhere in the body. Name normalisation cannot repair this, because no call carries code to normalise.
3. The remedy that works at evaluation time — forced tool choice or schema-constrained decoding — is **not exposed** by verl 0.9.0’s vLLM rollout path. Its only `strict` key belongs to the profiler.

This is a property of the training stack’s *configuration surface*, not a proof that FC is irreparable in principle. We previously named the missing capability as guided decoding; §4.8 corrects that. The repaired arm needs a parser registered in verl’s own registry that accepts the model’s actual output format — a code change rather than a different backend — and must be run at 7B under parameter-efficient tuning, for the reasons given there.

## D.5 The four obstacles to a repaired-FC training arm

We attempted to construct a repaired-FC RL condition and failed at four successive layers. Each attempt is a measurement, so we report them; three of the four dissolved on inspection and the corrected account is in §4.9.

(i) **A vLLM parser plugin does not reach the training loop.** verl’s AgentLoop performs its own tool-call parsing (`verl/experimental/agent_loop/tool_parser.py`, registered as `hermes`, adapted from vLLM v0.9.1). `multi_turn.format=hermes` selects *that* parser, not the server’s, so attaching a plugin to the vLLM rollout server changes nothing in training — a distinction worth stating because the two look identical from the outside.

(ii) **A verl-side parser recovers calls, but none of them are the tool.** We implemented a `qwen2_5_coder` parser inside verl’s registry accepting `<tools>` tags, `json` fenced blocks and bare JSON. On the exact training condition (1.5B, mandatory prompt,  $n=100$ ) it lifts recoverable calls from **0/100 (hermes) to 52/100**. Of those 52, **zero name run\_tests** (Table 27). Every call targets the function the task asks the model to *write* — `can_form_word({"tiles", "word"})`, `check_password_strength({"password"})` — the same signature as Llama’s failure in §4.5, now at a  $5\times$  smaller model. **Name normalisation cannot repair this: no call carries code to normalise.**

(iii) **A role-disambiguation few-shot makes it worse, not better.** Since the failure is semantic rather than syntactic, we targeted it directly: a system prompt containing the wrong call and the right call side by side, with an explicit warning that the task function is not a tool. Same model, same 100 items, same server; only the system prompt changes. The intervention made the model *more fluent at emitting the wrong call* ( $52 \rightarrow 64$  recoverable) and slightly worse at the task ( $15 \rightarrow 13$ ), with the count that matters staying at zero (Table 28). This is the third independent instance of the pattern in §G.9.

(iv) **Constrained decoding is unavailable in this stack.** Forcing the tool choice, or schema-constrained decoding, is not exposed by verl 0.9.0’s vLLM rollout path (its only `strict` key belongs to the profiler). **This obstacle is real but was the wrong diagnosis of what a repaired-FC arm needs:** guided decoding *forces* a format, whereas the repair this paper argues for is to accept what the model already emits. See §4.9.

## E Arm inventory

Every full-length arm, generated from the trajectory files’ own provenance rather than transcribed by hand. All 50 arms share one item set (`clean[:100]`; distinct-set count verified = 1), so every cross-arm comparison in the paper is exactly paired. Fields recorded as `---` were absent from that generation’s provenance schema (Appendix A.4); this is disclosed rather than back-filled. Seven arms carry a known-wrong `max_tokens` record and are listed in Appendix A.1; five are inadmissible for pass rates and are listed in Appendix A.3.

Not listed: `v8_Llama8B_recheck` (23 items, a targeted qualitative re-run) and eight `*smoke*` files at  $n=3$ . Neither is a formal arm.

Table 6: All 50 full-length arms and their recorded configuration.

| Arm                                | Model                 | Prot. | Strength | Schema | Adapter      | tok  | temp | seed |
|------------------------------------|-----------------------|-------|----------|--------|--------------|------|------|------|
| a14_Qwen1.5B_fc_mandatory          | Qwen-1.5B             | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| roleprobe_Qwen1.5B_fc_roledisambig | Qwen-1.5B             | fc    | optional | terse  | cross_family | 2048 | 0.0  | 0    |
| v11_DS1.3b_react                   | DS-1.3B               | react | optional | terse  | cross_family | 2048 | —    | —    |
| v11_DS6.7b_react                   | DS-6.7B               | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v11_Llama32_1B_fc_strict           | Llama-3.2-1B          | fc    | optional | strict | cross_family | 2048 | 0.0  | —    |
| v11_Llama32_1B_react               | Llama-3.2-1B          | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v11_Llama32_3B_fc_strict           | Llama-3.2-3B          | fc    | optional | strict | cross_family | 2048 | 0.0  | —    |
| v11_Llama32_3B_react               | Llama-3.2-3B          | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v13_DS1.3b_react                   | DS-1.3B               | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v13_DS6.7b_react                   | DS-6.7B               | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v13_Llama32_1B_react               | Llama-3.2-1B          | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v13_Llama32_3B_react               | Llama-3.2-3B          | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v13_Llama8B_fcstrict_t06_s1        | Llama-3.1-8B-Instruct | fc    | optional | strict | cross_family | 2048 | 0.6  | 1    |
| v13_Llama8B_fcstrict_t06_s2        | Llama-3.1-8B-Instruct | fc    | optional | strict | cross_family | 2048 | 0.6  | 2    |
| v13_Llama8B_fcstrict_t06_s3        | Llama-3.1-8B-Instruct | fc    | optional | strict | cross_family | 2048 | 0.6  | 3    |
| v13_Llama8B_react_t06_s1           | Llama-3.1-8B-Instruct | react | optional | terse  | cross_family | 2048 | 0.6  | 1    |
| v13_Llama8B_react_t06_s2           | Llama-3.1-8B-Instruct | react | optional | terse  | cross_family | 2048 | 0.6  | 2    |
| v13_Llama8B_react_t06_s3           | Llama-3.1-8B-Instruct | react | optional | terse  | cross_family | 2048 | 0.6  | 3    |
| v14_Llama8B_react                  | Llama-3.1-8B-Instruct | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v14_Mistral7B_react                | Mistral-7B            | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v14_Qwen7B_react                   | Qwen-7B               | react | optional | terse  | cross_family | 2048 | 0.0  | —    |
| v3_Llama8B_fc                      | Llama-3.1-8B-Instruct | fc    | optional | —      | cross_family | 2048 | —    | —    |
| v3_Llama8B_fc_legacy               | Llama-3.1-8B-Instruct | fc    | optional | —      | cross_family | 2048 | —    | —    |
| v3_Llama8B_react                   | Llama-3.1-8B-Instruct | react | optional | —      | cross_family | 2048 | —    | —    |
| v3_Mistral7B_fc                    | Mistral-7B            | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v3_Mistral7B_react                 | Mistral-7B            | react | optional | terse  | cross_family | 2048 | —    | —    |
| v3_Qwen7B_fc_legacy                | Qwen-7B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v3_Qwen7B_fc_nosuffix              | Qwen-7B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v3_Qwen7B_react                    | Qwen-7B               | react | optional | terse  | cross_family | 2048 | —    | —    |
| v4_Llama8B_fc_cot                  | Llama-3.1-8B-Instruct | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v4_Llama8B_fc_rich                 | Llama-3.1-8B-Instruct | fc    | optional | rich   | cross_family | 2048 | —    | —    |
| v5_Llama8B_fc_official             | Llama-3.1-8B-Instruct | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Mistral7B_fc_official           | Mistral-7B            | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Qwen1.5B_fc_intent              | Qwen-1.5B             | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Qwen14B_fc_intent               | Qwen-14B              | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Qwen32B_fc_intent               | Qwen-32B              | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Qwen3B_fc_intent                | Qwen-3B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v5_Qwen7B_fc_intent                | Qwen-7B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v6_Llama8B_fc_strict               | Llama-3.1-8B-Instruct | fc    | optional | strict | cross_family | 2048 | —    | —    |
| v6b_Qwen1.5B_fc_plugin             | Qwen-1.5B             | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v6b_Qwen3B_fc_plugin               | Qwen-3B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v6b_Qwen7B_fc_plugin               | Qwen-7B               | fc    | optional | terse  | cross_family | 2048 | —    | —    |
| v9_Mistral7B_fc_official_strict    | Mistral-7B            | fc    | optional | strict | cross_family | 2048 | —    | —    |
| v9_Mistral7B_fc_strict             | Mistral-7B            | fc    | optional | strict | cross_family | 2048 | —    | —    |
| x2_DS1.3B_react                    | DS-1.3B               | react | optional | —      | —            | —    | —    | —    |
| x2_DS6.7B_react                    | DS-6.7B               | react | optional | —      | —            | —    | —    | —    |
| x2_Llama8B_fc                      | Llama-3.1-8B-Instruct | fc    | optional | —      | —            | —    | —    | —    |
| x2_Llama8B_react                   | Llama-3.1-8B-Instruct | react | optional | —      | —            | —    | —    | —    |
| x2_Mistral7B_fc                    | Mistral-7B            | fc    | optional | —      | —            | —    | —    | —    |
| x2_Mistral7B_react                 | Mistral-7B            | react | optional | —      | —            | —    | —    | —    |

## F Human validation of the intent criterion, in full

The scale trend in §4.3 rests on an automated classifier, so we validated it against human judgement. 98 outputs were sampled **stratified by classifier verdict** (40 **tight**, 28 **strong-but-not-tight**, 30 **neither**) — deliberately over-sampling the decision boundary so both false positives and false negatives are estimable. Sampling is seeded and the annotator saw only the raw output, never the classifier’s label. We report **two rounds**, because the first is itself a finding.

Table 7: Human validation of the intent criterion, round 1 and its adjudication. The classifier is used only to count what the server discarded, so its recall matters more than its precision.

|                           | agreement    | Cohen’s $\kappa$ | precision    | recall | FP | FN |
|---------------------------|--------------|------------------|--------------|--------|----|----|
| Round 1 (raw output only) | 86.7%        | 0.713            | 70.0%        | 96.6%  | 12 | 1  |
| After adjudication        | <b>96.9%</b> | <b>0.936</b>     | <b>95.0%</b> | 97.4%  | 2  | 1  |

The 12 round-1 false positives split into two causes, and the second is ours, not the annotator’s. **Eleven are reading-order failures:** the tool call sits mid-document, after a fenced `python` block (match positions 1209–2585 in outputs of median length  $\sim 2000$ ); the annotator read the code block, concluded “this is a direct answer, not a call”, and stopped. **One is an instrument failure:** for a 32B item the matched span begins at character 2585 and the annotation pack truncated every output at 2400, so the evidence was *not in the pack at all* and that item could not have been labelled correctly. Re-shown the matched span alone, 10 of 13 disputed items were reversed. The corrected pack carries every output in full.

**The first adjudication was one-sided, and we do not treat its  $\kappa$  as a reliability estimate.** Auditing the procedure after the fact: the 13 adjudicated items were selected *exactly* as the items where the annotator disagreed with the classifier (13/13 overlap), and all 10 reversals moved *toward* the classifier, none away. Under that selection rule agreement can only rise, so  $0.713 \rightarrow 0.936$  is a property of which items were re-examined, not evidence of reliability. We report the second annotator instead.

### A second independent annotator

A second annotator, blind to the classifier, to the first annotator’s labels and to the first pack’s item order, labelled the identical 98 items under a verbatim-identical rubric. Two changes were made to the instrument: outputs are carried **in full**, and the adjudication rule was fixed in advance to review *all* three-way disagreements rather than only those disagreeing with the classifier.

Table 8: Inter-annotator reliability. Restricting to items where both annotators read identical bytes is not a convenience: 11 of the 12 raw disagreements are items the first pack truncated, where A1 answered “no call” because the call had been cut from their copy.

| Cohen’s $\kappa$ (95% CI, bootstrap) | all items ( $n=97$ ) | <b>identical bytes (<math>n=62</math>)</b> |
|--------------------------------------|----------------------|--------------------------------------------|
| A1 vs. A2 (inter-annotator)          | 0.736 [0.590, 0.863] | <b>0.967 [0.897, 1.000]</b>                |
| A2 vs. classifier                    | 0.936 [0.855, 1.000] | <b>0.967 [0.897, 1.000]</b>                |
| A1 vs. classifier                    | 0.712 [0.559, 0.847] | 0.934 [0.834, 1.000]                       |

**The apparent unreliability was the instrument, not the raters.** Of the 12 items where the two annotators disagree, **11 have full text exceeding 2400 characters**, and in every one



A1 answered N while A2 answered Y. On the 62 items where both read identical bytes the two agree on 61, and **inter-annotator**  $\kappa = 0.967$ .

**Third-party adjudication, and the final labels.** Fifteen items were not unanimous across A1, A2 and the classifier. All fifteen went to a third adjudicator under the pre-registered rule — selected by *any* three-way disagreement, never by disagreement with the classifier — who saw full text and no one else’s labels. The adjudicator sided with A2 on 11, with the classifier on 9, and with A1 on 6, and **moved away from the classifier on 6 of 15**. That last number is the check that matters: the first round’s adjudication moved away from the classifier zero times out of thirteen, which is what an outcome looks like when the selection rule has already decided it. Combining the 83 unanimous items with the 15 adjudicated ones gives a gold standard that at no point consulted the classifier.

Table 9: Validation against the adjudicated gold standard. The inter-annotator column is restricted to items where both annotators read identical bytes; the classifier and the gold standard are unaffected by that truncation and are scored on the full sample.

| Cohen’s $\kappa$ (95% CI, bootstrap) | full sample ( $n=97$ )      | identical bytes ( $n=62$ )  |
|--------------------------------------|-----------------------------|-----------------------------|
| A1 vs. A2 (inter-annotator)          | 0.736 [0.595, 0.866]        | <b>0.967 [0.893, 1.000]</b> |
| Classifier vs. gold standard         | <b>0.871 [0.756, 0.958]</b> | 0.934 [0.829, 1.000]        |

**Correction factor.** Against the gold standard the **tight** stratum’s precision is  $36/40 = 0.900$ , so §4.3’s headline 80/100 at 32B corrects to  $\approx 72$ . The criterion errs in both directions — 4 over-counts against 2 calls it misses entirely, a net over-count of 2 — so the correction is smaller than the precision figure alone suggests. **The trend is unaffected:** applying 0.900 uniformly gives 0, 3.6, 18.9, 32.4, 72 against a server-side zero at every size. We report uncorrected classifier counts in all tables with this factor stated, rather than rescaling silently.

For the record, this number moved twice as the validation improved, and we report the trajectory rather than only its endpoint: **0.950** under the first, one-sided adjudication; **0.975** under A2’s labels alone; **0.900** against the adjudicated gold standard. The first was inflated by its selection rule, the second rested on a single rater, and only the third is built from a rule fixed before the labels were seen.

**A bound on the measurement itself.** The probe stores at most 4000 characters of raw output per item; 10 of the 98 sampled outputs reach that ceiling, and one grey-zone item is a call truncated by it. The classifier and both annotators read identical bytes, which is what agreement requires, but a call emitted beyond character 4000 is invisible to all three. Emitted-call counts are accordingly a **lower bound** — conservative for our claim, since it can only understate the censoring we report.

## G Supplementary results

### G.1 The anomaly that motivated the study

### G.2 Result tables referenced from §4

Table 10: BFCL v4, first HTTP request per case. Intent columns are measured only where the parse failed.

RL improves first drafts; multi-turn repair never moves

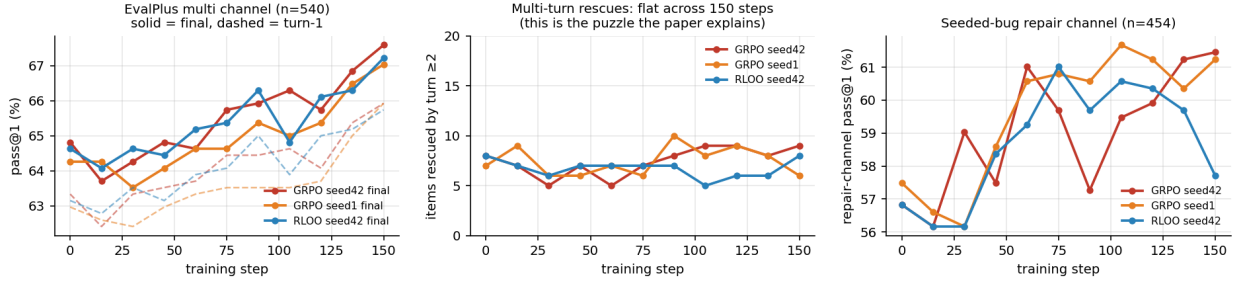


Figure 2: Left: pass@1 rises over 150 steps on all three runs, driven by first drafts (dashed) rather than final answers. **Centre: the count of items rescued by turn 2 or later never moves** — 6–9 out of 540 from step 0 to step 150, across two algorithms and two seeds. Right: the seeded-bug repair channel does improve, confirming the model is learning something about fixing code; it simply never does so *across turns*. §4.8 shows the tool was never successfully called in any of these runs.

| Arm                      | cases | HTTP err | first parsed | unparsed | tight | strong | weak |
|--------------------------|-------|----------|--------------|----------|-------|--------|------|
| hermes (documented)      | 200   | 0        | <b>0</b>     | 200      | 166   | 169    | 187  |
| qwen2_5_coder (repaired) | 200   | 0        | <b>196</b>   | 4        | 0     | 1      | 2    |

Table 11:  $\tau$ -bench **retail**, all 115 tasks paired across arms (none dropped).

|                                         | documented | repaired   |
|-----------------------------------------|------------|------------|
| assistant turns                         | 1389       | 1971       |
| server-parsed calls                     | <b>0</b>   | <b>636</b> |
| emitted but unparsed                    | <b>789</b> | 6          |
| tool executions / observations returned | <b>0</b>   | <b>636</b> |
| tasks entering the tool loop at all     | <b>0</b>   | <b>103</b> |
| tasks solved                            | 7          | 10         |

Table 12: Same probe, items and criterion under a *matched* envelope.

| Size                                | server-parsed   | silent fraction (tight, unparsed)                                                                       |
|-------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------|
| Qwen3-0.6B                          | 86              | <b>1</b>                                                                                                |
| Qwen3-1.7B                          | 44              | <b>1</b>                                                                                                |
| Qwen3-4B                            | 70              | <b>2</b>                                                                                                |
| Qwen3-8B                            | 97              | <b>0</b>                                                                                                |
| Qwen2.5-Coder 1.5B–32B (mismatched) | 0 at every size | <b>0</b> $\rightarrow$ <b>4</b> $\rightarrow$ <b>21</b> $\rightarrow$ <b>36</b> $\rightarrow$ <b>80</b> |

Table 13: Qwen2.5-Instruct under the identical mismatched configuration, with both ladders’ task outcomes measured on a working executor.

| Size       | Instruct, emitted |              |      | Coder  |                | Instruct |       |                |
|------------|-------------------|--------------|------|--------|----------------|----------|-------|----------------|
|            | parsed            | <b>tight</b> | weak | parsed | turn-1 = final | turn-1   | final | <b>rescued</b> |
| 1.5B       | 1                 | 0            | 7    | 0      | 31             | 13       | 13    | <b>0</b>       |
| 3B         | 11                | 56           | 2    | 0      | 37             | 14       | 17    | <b>+3</b>      |
| 7B         | 63                | 21           | 0    | 0      | 53             | 34       | 40    | <b>+6</b>      |
| 14B        | 77                | 19           | 0    | 0      | 53             | 38       | 52    | <b>+14</b>     |
| <b>32B</b> | <b>88</b>         | 8            | 0    | 0      | 68             | 42       | 65    | <b>+23</b>     |

Table 14: Llama-3.1-8B arms, paired on the same 100 items.

| Arm                                 | Calls issued | Wrong-tool | Executed  | Turn-1    | Final     |
|-------------------------------------|--------------|------------|-----------|-----------|-----------|
| terse (parser only)                 | 97           | 23         | 74        | 34        | 49        |
| + official template                 | 95           | 22         | 73        | 33        | 44        |
| + official template + <b>strict</b> | 98           | <b>0</b>   | <b>97</b> | <b>46</b> | <b>61</b> |
| ReAct (reference)                   | 98           | 0          | 97        | <b>61</b> | 80        |

Table 15: Decomposing the ReAct–FC gap on Llama, paired McNemar.

| Comparison                     | b/c  | <i>p</i>      | Attributable to              |
|--------------------------------|------|---------------|------------------------------|
| terse-FC → strict-FC (49 → 61) | 15/3 | <b>0.0075</b> | <b>interface repair, +12</b> |
| strict-FC → ReAct (61 → 80)    | 27/8 | <b>0.0019</b> | <b>protocol, +19</b>         |
| terse-FC → ReAct (49 → 80)     | 38/7 | 3.1e-06       | both, +31                    |

Table 16: Repair loop on Qwen2.5-Coder-7B, `tool_choice: auto` fixed, changing only the adapter.

| Arm                    | Parsed    | Mean turns | $\geq 2$ turns | Turn-1 | Final     | Rescued  |
|------------------------|-----------|------------|----------------|--------|-----------|----------|
| ReAct                  | 95        | 1.92       | 40             | 57     | <b>74</b> | 17       |
| FC + hermes            | <b>0</b>  | 0.96       | <b>0</b>       | 53     | 53        | <b>0</b> |
| FC + dedicated adapter | <b>84</b> | 1.82       | <b>37</b>      | 53     | <b>62</b> | <b>9</b> |

Table 17: RL training rollouts, matched in model, data, hyper-parameters, seed and prompt strength.

| Arm                                    | Steps | num_turns/mean          | Tool time     | Tool calls / rollout |
|----------------------------------------|-------|-------------------------|---------------|----------------------|
| FC ( <code>tool_agent</code> + hermes) | 10    | <b>2.000</b> (constant) | <b>0.00 s</b> | <b>0.000</b>         |
| ReAct ( <code>react_agent</code> )     | 3     | 5.883                   | 12.21 s       | <b>2.052</b>         |

### G.3 Offline re-parse matrix and failure-layer decomposition

To exclude “your extractor is simply better than **hermes**”, we re-parsed the *same stored bytes* under four rules (`analysis/reparse_matrix.py`, no GPU).

Table 18: The same bytes, re-parsed offline under four rules. Where the server already parsed a call, **content** is empty and offline replay has nothing to read; those cells are N/A rather than zero. The gap is a parser mismatch, not a claim that no parser could have read the output.

| Arm (server-parsed = 0) | server | hermes replay | <tools> replay | bare JSON  | tight     |
|-------------------------|--------|---------------|----------------|------------|-----------|
| Qwen-1.5B               | 0      | 0             | 0              | 0          | 0         |
| Qwen-3B                 | 0      | 0             | 0              | 5          | 4         |
| Qwen-7B                 | 0      | 0             | 0              | 22         | 21        |
| Qwen-14B                | 0      | 0             | 0              | 42         | 36        |
| Qwen-32B                | 0      | <b>0</b>      | 0              | <b>100</b> | <b>80</b> |

The **hermes** column is zero at every scale: the model never produces that format. The bare-JSON and tight columns are not: the calls exist, and their difference from the server column is precisely what the format mismatch removes. For arms where the server *did* parse, offline content-only re-parsing is **undefined, not zero** — a successful parse moves the call into `tool_calls` and leaves `content` empty.

Replaying vLLM 0.27.1’s **hermes** extractor line for line over the stored bytes (`analysis/failure_layer.py`) separates the layer at which each item is lost: no `<tool_call>` envelope at all; envelope present but the JSON payload malformed; payload well-formed and rejected only because `json.loads` does not tolerate trailing text in the capture; or a genuine parser loss, meaning the replay succeeds where the server did not.

Table 19: Where the unparsed items are lost. Coder fails entirely at the envelope; Instruct reaches the envelope and fails at the payload. **Genuine parser loss is zero in every arm** — and across all 4254 de-duplicated first turns in the archive.

| Arm                        | no envelope | payload malformed | strictness only | parser loss |
|----------------------------|-------------|-------------------|-----------------|-------------|
| Coder 1.5B–32B (all sizes) | 100         | 0                 | 0               | <b>0</b>    |
| Instruct 1.5B              | 93          | 6                 | 0               | <b>0</b>    |
| Instruct 3B                | 22          | 41                | 26              | <b>0</b>    |
| Instruct 7B                | 3           | 33                | 1               | <b>0</b>    |
| Instruct 14B               | 13          | 6                 | 4               | <b>0</b>    |
| Instruct 32B               | 1           | 9                 | 2               | <b>0</b>    |

The two ladders fail at different layers: Coder emits a well-formed bare call and never the envelope, so **hermes** not seeing it is *correct behaviour*; Instruct emits the envelope and then breaks the JSON, almost always by embedding Python whose docstring quotes or raw newlines are left unescaped inside the code string. And **in no arm** — **across the whole archive** — **did the server fail to parse a call that its own extractor would have accepted**. “Censoring” throughout this paper therefore means *the attempt never reaches the parser in a form it is defined to accept*, not *the parser discards valid calls*.

#### G.4 BFCL: gating and the full reproduction record

Each arm passes a two-case end-to-end smoke test before its full run, and a run is rejected unless the result count, the score file’s `total_count` and the number of distinct case ids in the recording proxy all agree at 200 with zero HTTP errors. Both arms run at `max_model_len = 16384`, `temperature = 0` and inference seed 0, on cases drawn at manifest seed 20260901. Registration of BFCL’s OpenAI handler happens in memory at import time, so `bfcl_eval`’s own files are left byte-identical; a recording proxy sits between benchmark and server and logs every request.

We ran the pair twice. The **hermes** arm is identical across runs in every column: 0/200 parsed, 166 tight, 480 requests, 0 errors. The repaired arm moves by one case between runs — **197 versus 196** first parsed, **99 versus 98** executed — while success is 19 in both. Greedy decoding does

not make vLLM bitwise deterministic across differing batch compositions, and this arm runs eight threads through a multi-turn executor, so we report its intermediate funnel columns as  $\pm 1$  rather than exact.

Table 20: BFCL `multi_turn_base`,  $n = 100$ , joined to the HTTP trace by case id.

| Arm                        | tight (unparsed) | first parsed | any parsed | executed | <b>success</b> | rescued |
|----------------------------|------------------|--------------|------------|----------|----------------|---------|
| <code>hermes</code>        | 69               | 0            | 0          | 0        | <b>0</b>       | 0       |
| <code>qwen2_5_coder</code> | 0                | 97           | 98         | 98       | <b>19</b>      | 0       |

Table 21: Two quantified failure layers; the full four-family table is Table 22.

| Family        | Layer  | Symptom                                                       | Detectable in ad-<br>vance?           | Remedy              |
|---------------|--------|---------------------------------------------------------------|---------------------------------------|---------------------|
| Qwen2.5-Coder | parser | emits a fenced JSON block, not <code>&lt;tool_call&gt;</code> | $\times$ template check               | dedicated adapter   |
| Llama-3.1-8B  | schema | calls the <i>task function</i> as a tool (23%)                | $\times$ requires argument inspection | <b>strict: true</b> |

## G.5 The full four-family taxonomy

Table 22: Four families, four layers. Only Mistral’s failure announces itself.

| Family          | Layer    | Symptom                                                       | Detectable in ad-<br>vance?           | Remedy              |
|-----------------|----------|---------------------------------------------------------------|---------------------------------------|---------------------|
| DeepSeek-Coder  | template | chat template never injects tools                             | $\checkmark$ template inspection      | none                |
| Qwen2.5-Coder   | parser   | emits a fenced JSON block, not <code>&lt;tool_call&gt;</code> | $\times$ template check               | dedicated adapter   |
| Llama-3.1-8B    | schema   | calls the <i>task function</i> as a tool (23%)                | $\times$ requires argument inspection | <b>strict: true</b> |
| Mistral-7B-v0.3 | token    | repeats <code>[TOOL_CALLS]</code> $\rightarrow$ HTTP 400      | $\sim$ errors, but only sometimes     | <b>none found</b>   |

## G.6 Main table under unified configuration

Table 23: All arms genuinely at `max_tokens=2048`, one shared 100-item set, `temperature=0`. Mistral has no admissible FC arm: all four configurations produce request errors, so no pass rate and no  $p$ -value are computed.

| Model            | ReAct  |           | FC     |       | b/c  | $p$           | FC arm                         |
|------------------|--------|-----------|--------|-------|------|---------------|--------------------------------|
|                  | turn-1 | final     | turn-1 | final |      |               |                                |
| Llama-3.1-8B     | 61     | <b>80</b> | 46     | 61    | 27/8 | <b>0.0019</b> | official tmpl. + <b>strict</b> |
| Qwen2.5-Coder-7B | 57     | <b>74</b> | 53     | 62    | 17/5 | <b>0.0169</b> | dedicated adapter              |
| Mistral-7B-v0.3  | 26     | 33        | N/A    | N/A   | N/A  | N/A           | —                              |

## G.7 The four Llama controls, tabulated

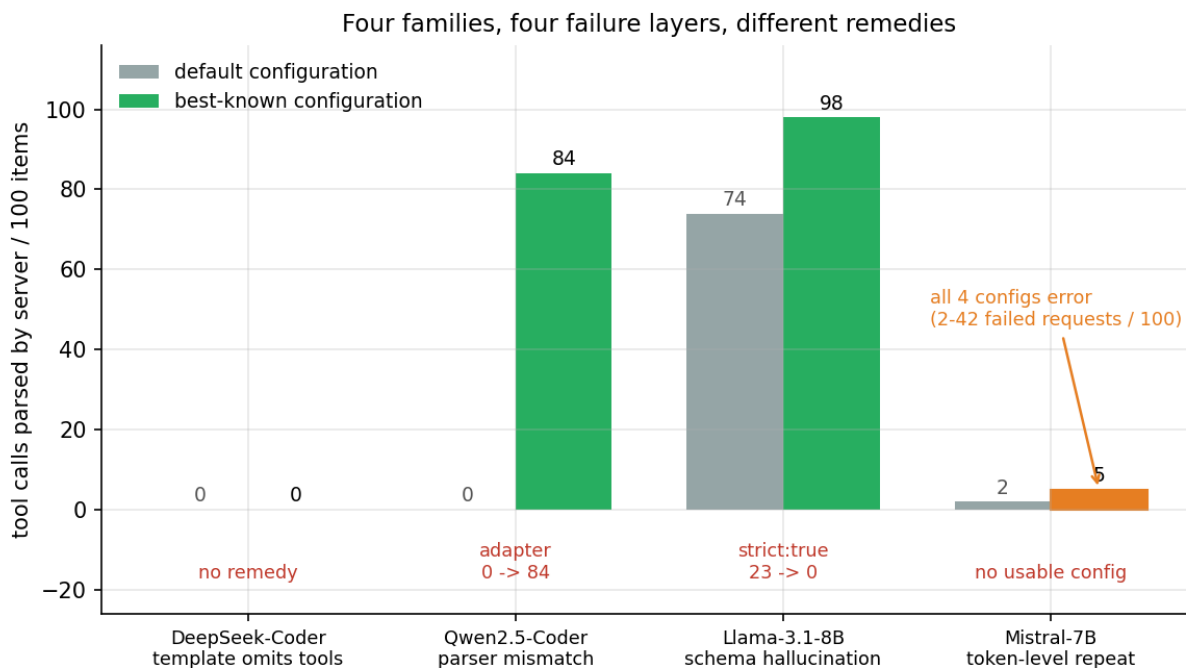


Figure 3: The same four failures, quantified. Grey is the configuration a reader arrives at by following the documentation; green is the best configuration we found. Mistral never produces an error-free run under any of four configurations, so its bars are not comparable with the others and are drawn in orange.

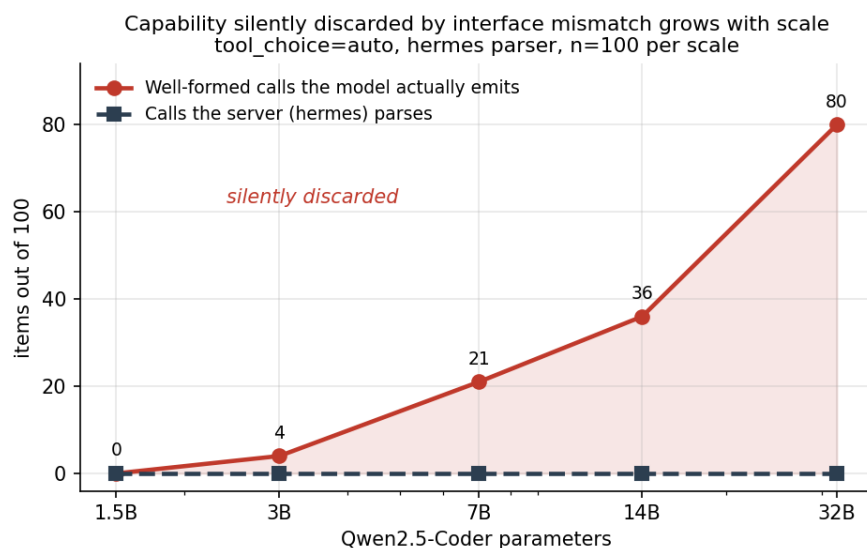


Figure 4: **The undercount grows with scale.** Across a  $21\times$  range the server parses zero calls at every size, while well-formed calls the model actually emits rise to 80/100 at 32B (72 after the correction factor of 0.900, Appendix F). Same data as Table 2.

Table 24: Four single-variable controls on Llama-3.1-8B’s 23% wrong-tool rate. The first three rejected their alternative explanations and pointed at the model; the fourth found the actual cause and withdrew that conclusion.

| Alternative explanation                 | Control                  | Empty-argument rate                                     | Verdict         |
|-----------------------------------------|--------------------------|---------------------------------------------------------|-----------------|
| schema lacks a parameter description    | rich vs terse            | 23 $\rightarrow$ 22 ( $p=1.000$ )                       | rejected        |
| ReAct merely adds a reasoning step      | FC + Thought scaffold    | 23 $\rightarrow$ <b>59</b> ( $p<0.001$ , <b>worse</b> ) | rejected        |
| server not configured per documentation | + official chat template | 23 $\rightarrow$ 22 ( $p=0.125$ )                       | rejected        |
| <b>schema constraint not enabled</b>    | <b>+ strict: true</b>    | <b>23 <math>\rightarrow</math> 0</b> ( $p=0.0001$ )     | <b>accepted</b> |

## G.8 Conditioning on a successful parse, and access vs. use of feedback

Table 25: Conditioning on items where both protocols parsed successfully. The residual protocol gap holds on Llama and does not reach significance on Qwen — the honest reading is one family out of two, not a general law.

| Family           | Items both parsed | ReAct | FC | b/c  | $p$                 |
|------------------|-------------------|-------|----|------|---------------------|
| Qwen2.5-Coder-7B | 83                | 63    | 56 | 11/4 | <b>0.119 (n.s.)</b> |
| Llama-3.1-8B     | 96                | 78    | 60 | 25/7 | <b>0.0021</b>       |

Table 26: Repair success given that feedback was actually received. Conditioned this way the two protocols are much closer, which locates the gap in *how often feedback arrives* rather than in what the model does with it. On Qwen this difference is within the non-significant band above, so we state it as an observation, not a result.

| Arm          | Reached turn $\geq 2$ | Rescued  | Repair success given feedback |
|--------------|-----------------------|----------|-------------------------------|
| ReAct        | 40                    | 17       | <b>42%</b>                    |
| FC + adapter | 37                    | 9        | <b>24%</b>                    |
| FC + hermes  | <b>0</b>              | <b>0</b> | undefined                     |

**Adapters are not a universal switch.** The same adapter recovers 84/100 at 7B and **1/100 at 3B**: the 3B checkpoint ignores the `<tools>` few-shot entirely (zero `<tools>` tags in 100 items, 99 direct code) while still solving the task at a comparable rate (final 53). Adapter effectiveness is non-monotonic in scale, reinforcing the per-checkpoint claim of §4.6. The dedicated solution also changes parser, chat template and few-shot simultaneously; we additionally rewrote the template’s few-shot examples, which originally invoked `get_weather` — a tool absent from our schema, which would have induced calls to a non-existent tool and inflated both initiation and empty-argument counts. Original and modified hashes are recorded.

## G.9 Pressure on a broken interface makes things worse

Two independent controls point the same way. Adding a reasoning scaffold to FC raised role confusion from 23 to **59** (§4.5). Replacing the optional tool instruction with a mandatory one on Qwen2.5-Coder-1.5B moved parser-accepted calls not at all — still 0/100 — while final pass fell from **31 to 15** and unparsable outputs rose from  $\sim 0$  to 46/100. Neither intervention touches the interface, and both make the observable outcome worse. **When an agent appears reluctant**



to use tools, strengthening the instruction is the cheapest thing to try and the most likely to mislead — it can degrade task performance while leaving the tool-call count at zero, which looks like a model that is both unwilling *and* incapable.

Table 27: Probe replication on the 1.5B model actually being trained, same mandatory prompt (n=100). Not one output names `run_tests`, so at this scale the missing calls are not a parser mismatch — there was nothing well-formed to censor.

| Category                                                        | n         | %   |
|-----------------------------------------------------------------|-----------|-----|
| wrong name, <b>arguments</b> carry complete code                | <b>0</b>  | 0%  |
| wrong name, <b>arguments</b> are the task function’s parameters | <b>36</b> | 69% |
| wrong name, executable code present elsewhere in the body       | 16        | 31% |

Table 28: Increasing pressure on the 1.5B model. Stronger instruction and few-shot examples raise the number of recoverable code payloads but never produce a call that names the tool.

|                                | recoverable calls | <b>naming run_tests</b> | final pass |
|--------------------------------|-------------------|-------------------------|------------|
| mandatory prompt (baseline)    | 52                | <b>0</b>                | 15         |
| + role-disambiguation few-shot | <b>64</b>         | <b>0</b>                | <b>13</b>  |

## G.10 The sufficiency arm, tabulated

RL training rollouts: under FC the tool is never successfully called  
Qwen2.5-Coder-1.5B, verl GRPO, mandatory tool-use prompt on both arms

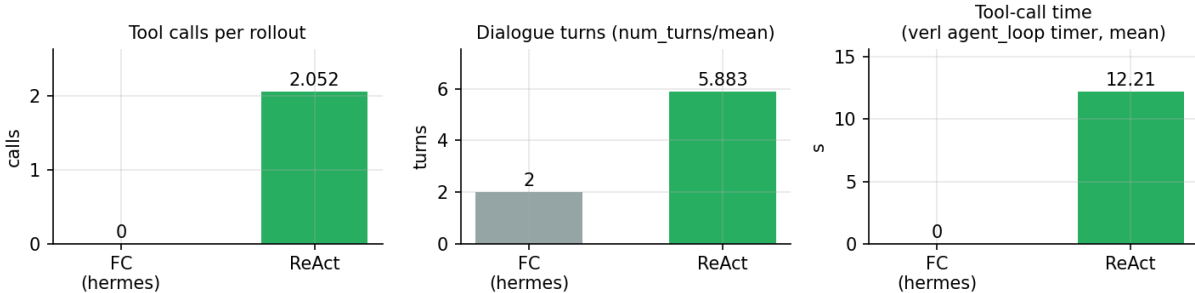


Figure 5: **RL training rollouts.** Under function calling the tool is never successfully called; `num_turns` stays pinned at its minimum of 2 and tool time is exactly zero. Prompt strength is matched across arms (both mandatory); step counts are not (10 vs 3).

Table 29: ReAct training with a working tool channel, 23,676 executed calls. Rescues by turn  $\geq 2$  do not move. The three historical FC runs, in which **zero** tool calls executed, give rescue sequences [7, 9, 6, 6, 7, 6, 10, 8, 9, 8, 6], [8, 7, 6, 7, 7, 7, 7, 5, 6, 6, 8] and [7, 5, 7, 5, 7, 8, 9, 9, 8, 9] — the two conditions are indistinguishable on this metric.

| step | turn-1 | final | rescued by turn $\geq 2$ | repair channel |
|------|--------|-------|--------------------------|----------------|
| 0    | 359    | 367   | 8                        | 263            |
| 15   | 340    | 346   | <b>6</b>                 | 266            |
| 30   | 340    | 346   | <b>6</b>                 | 262            |
| 45   | 341    | 349   | <b>8</b>                 | 261            |
| 60   | 341    | 347   | <b>6</b>                 | 274            |

### G.11 Sampling variance, and an inverted noise structure

Llama-3.1-8B, temperature=0.6, n=100  $\times$  3 seeds.

Table 30: Sampling variance. FC seed 3 is inadmissible (a parallel tool call vLLM rejects), leaving two arms — too few for a standard deviation, so only the interval is reported.

|              | seed 1 | seed 2 | seed 3        | admissible |
|--------------|--------|--------|---------------|------------|
| ReAct turn-1 | 53     | 56     | 43            | 3/3        |
| ReAct final  | 72     | 72     | 72            | 3/3        |
| FC turn-1    | 44     | 44     | 45            | —          |
| FC final     | 62     | 66     | <del>57</del> | <b>2/3</b> |

FC seed 3 contains one request error — the model emitted parallel tool calls and vLLM returned “*This model only supports single tool-calls at once*” — so it is inadmissible by §3.3. With two admissible FC arms we report the interval [62, 66] and **no standard deviation**. Worst case: ReAct’s lowest (72) versus FC’s highest (66) = **+6**; the direction is consistent across all admissible repetitions.

**Inverted noise structure.** ReAct’s turn-1 varies widely (43–56, sd 6.8) while its final converges; FC’s turn-1 is nearly constant (44–45, sd 0.58) while its final disperses. The mechanism is compensation in the rescue count — 19 / 16 / **29**, largest where turn-1 was worst — and in all three seeds the count of “passed at turn 1 but failed finally” is **0**: the repair loop is monotone. **The three identical finals are a coincidence of totals, not identical item sets**, and sd = 0.00 must not be read as “ReAct is perfectly stable”.

### G.12 Multi-turn repair stabilises the total while its composition keeps moving

Two independent observations show the same structure. *Across sampling seeds*: ReAct’s final pass is 72 / 72 / 72 at temperature=0.6, yet the solved sets have pairwise Jaccard 0.71–0.80 — 15–24 items differ. *Across training steps*: between steps 15 and 30 of the ReAct run, 85 of 542 first-turn completions changed verbatim and 6 items flipped outcome (3 gained, 3 lost), leaving turn-1 and final pass numerically identical (340 / 346 at both checkpoints). The policy is demonstrably moving; the aggregate is not.

The common structure is that **the repair loop absorbs perturbation upstream of it**. Two consequences follow. *For measurement*, an aggregate pass rate is a **poor instrument for detecting change in a multi-turn agent**: it is stabilised by the very mechanism under study, and item-level set comparison detects movement the total conceals. We report both throughout, and note that had we reported only totals we would have concluded — twice, on two different axes — that nothing was happening. *For deployment*, stability of the total under sampling noise is a property practitioners want, it is contributed by the multi-turn loop rather than the base policy, and it is invisible in single-turn evaluation. **Bound**: both observations are on one model family

and one task, the identical totals are in part coincidence, and we do not claim the aggregate is invariant — only that it is far less sensitive than its components.

### G.13 The training verifier: seams, audit, and planned fixes

The GRPO reward is outcome-only: it extracts the last submitted program, runs the item’s own `pytest` suite in the sandbox, and returns the pass ratio (`reward_code.py`). Three reward-hacking routes are closed by construction — a run in which no test executes scores 0 rather than 1, a timeout scores 0, and an item with no test string scores 0. Two seams remain. (i) The summary parser searches `(\d+) passed` over the merged stdout/stderr buffer and takes the first match, so a program that itself prints a `pytest`-like line could be scored on its own output — this needs no understanding of the harness and could fire by accident. (ii) `skipped` and `xfail` are not counted in the denominator, so a rollout that skips the hard tests is scored on the subset it did not skip. Solutions execute at import time under the double-file layout, so both are reachable in principle. We therefore audited: across the five training logs that retain generated code there are **zero** occurrences of a printed ```N passed```, of `pytest.skip/allow_module_level`, or of `sys.exit/os._exit`, and `critic/rewards/mean` oscillates in 0.46–0.73 without the step change a discovered exploit produces. **The audit’s own limit is that full rollout text was not retained**, so it covers the code fragments those logs happen to carry, not the entire corpus. Neither seam can reach the reported pass rates, which use a stricter path (`mode="script"`, exit code only, `all_passed` additionally requiring `status=="ok"`). Before further training we would anchor the regex to `pytest`’s own summary line, count `skipped` and `xfail`, and separate model stdout from harness output; we do not change it now because doing so would make future runs incomparable to those reported here.

## H Limitations, in full

In the order a sceptical reader should apply them.

**Scope of the evidence.** (1) *One task family for the scale and training results.* Our own probe uses decontaminated KodCode items with a single tool taking one `code` argument. The external-validity results are *not* code-only: BFCL’s `multi_turn_base` spans file-system, vehicle and trading APIs, and  $\tau$ -bench `retail` is a customer-service domain whose 789 unparsed turns decompose to 100% envelope-layer with zero parser loss, reproducing the code result (§4.2). What remains code-only is the *scale ladder* and the *RL contamination* result. The former is a real gap — the same ladder on a non-code task would test whether the undercount’s growth with checkpoint size is a property of the interface or of code-shaped payloads (§6). The latter we do not expect to close: the reward is a `pytest` pass ratio, so porting it changes environment, reward and task at once. Multi-tool schemas and tool selection among alternatives remain untested throughout; a single-tool schema is the *easiest* case for a parser, which if anything makes the measured censoring a lower bound — but that is an argument, not a measurement. (2) *Item sampling.* The 50 full-length arms use 100 items taken as `clean[:100]`, **not a random sample**; pairing across arms is exact, so within-study comparisons are sound. Generalisation rests on a replication on a random 300-item sample (seed 20260901): server-parsed remains 0 at every size across 1500 items, and strict emitted counts scale to 0, 4, 30, 40, 81 per 100 against the original 0, 4, 21, 36, 80. **7B moves from 21 to 30**, so the original subset understated that rung; direction and monotonicity are unaffected, but a single 100-item subset is evidently not a stable estimate of an intermediate rung. (3) *The scale result is within one family.* The  $0 \rightarrow 4 \rightarrow 21 \rightarrow 36 \rightarrow 80$  ladder is Qwen2.5-Coder alone, against one mismatched interface, on one task; we do not claim censoring is generally scale-increasing, and we do not equate parameter count with capability. The Instruct ladder is a control on the *mechanism*, not a second

ladder of undercount — its execution layer was initially void (the host lacked `pytest`) and has since been re-run with a working executor, the parse layer reproducing value for value. A second *family’s* ladder remains the test that would promote or refute the scale claim. (4) *The taxonomy is four instances, not an enumeration.* The evidence is deliberately asymmetric: Qwen has a scale ladder, a replay matrix and a repair; Llama a single-variable intervention; **Mistral has no admissible 100-item FC comparison at all**; DeepSeek is a template check with little quantification. (5) *Single-sample headline arms.* Table 16 and the ladders are one sample at `temperature=0`. Variance is estimated only on Llama-3.1-8B, and there only 2 of 3 FC arms are admissible — too few for a standard deviation (§G.11). (6) *Multiple comparisons.* Roughly a dozen McNemar tests; the Bonferroni threshold at that family size is  $\approx 0.0042$ . Surviving it: the `strict: true` intervention ( $p=0.0001$ ), the protocol gap on 300 random items ( $p=3.1\times 10^{-6}$ ), and the main effects of order  $10^{-3}$ . **Not surviving it:** the repair-loop pass gain ( $p=0.093$  at  $n=100$ ,  $p=0.0385$  at  $n=300$ ), the Llama protocol contrast at  $n=300$  ( $p=0.0352$ ), and the residual-gap estimates ( $p=0.118$ ,  $0.125$ ). We rest no claim on the latter group, and report the  $n=300$  values precisely so a reader can see the direction is stable where significance is not claimed. (7) *Model scale stops at 32B*, on a single A800; no frontier models.

**Threats to the causal reading.** (8) *There is no clean repaired-FC training control.* The training evidence compares FC against ReAct, changing protocol and interface together; ReAct is a positive-control interaction channel, not a parser-repair control. We previously attributed the arm’s absence to verl 0.9.0 lacking guided decoding. **That was wrong:** the registry is extensible and we have since installed code in it, and what actually blocks the arm is scale (§4.9). Consequently we claim that interface censoring can be observed directly inside rollout collection and eliminates tool-mediated samples; we do **not** claim to have causally proven that it prevents RL from learning multi-turn repair. (9) *Conditioning on a successful parse is post-treatment conditioning.* Whether an item parses is determined by the interface under test, so the both-parsed subset is not a random subsample and its contrast is not a clean causal estimate. On Qwen the residual is +8.4 pp, 95% CI  $[-0.5, +17.4]$ ,  $p=0.118$ : **we do not detect a residual difference, which is not the same as establishing there is none.** (10) *Observed zero is not zero probability.* Our instrumentation records zero parser-accepted tool executions across every step of the FC arm — a statement about the sampled experience distribution, not the support of the policy, since entropy and off-branch updates could in principle carry the model into parser-compatible output. We therefore say the branch receives no direct on-policy gradient signal under the observed rollout distribution and is effectively inaccessible in this cold-start regime, not mathematically unreachable. (11) *Zero executions is not zero attempts.* The FC training arm did not save raw rollout text, so we cannot separate “never attempted” from “attempted and discarded”. The probe replication at the same scale and prompt finds no output naming `run_tests`, which makes “nothing well-formed to censor” the most likely reading at 1.5B — but it is a replication, not the training run itself. Retaining rollout text is a change we would make before any further training. (12) *Opening the channel is tested only through ReAct.* The 75-step arm executing 23,676 calls shows a working channel does not by itself move multi-turn rescues, but because it is not a parser-repair control it cannot separate “a working channel is insufficient” from “ReAct in particular is insufficient”, and it is 75 steps on one seed at 1.5B against 150-step historical baselines. (13) *Human validation covers one criterion on one sample.* Two blind annotators on 98 stratified items give inter-annotator  $\kappa = 0.967$  on the 62 items where both read identical bytes, and classifier-vs-gold-standard  $\kappa = 0.871$  after third-party adjudication (Appendix F). Two residual limits: the sample is stratified by classifier verdict, so it estimates per-stratum precision and recall, not a population rate; and stored raw output is capped at 4000 characters, so a call emitted past that point is invisible to classifier and annotators alike, making emitted-call counts a lower bound. The *first* round’s apparent unreliability

( $\kappa=0.713$ ) was an artefact of a 2400-character truncation in the annotation pack, not of the raters; that pack is superseded. (14) *The training verifier has two exploitable seams, which we audited rather than assumed away.* The GRPO reward is outcome-only — last submitted program, item’s own `pytest` suite, pass ratio — and closes three reward-hacking routes by construction (no test executed, timeout, and no test string all score 0). Two seams remain: the summary parser takes the first `(\d+)` `passed` match over merged `stdout/stderr`, so a program printing a `pytest`-like line could be scored on its own output; and `skipped/xfail` are not in the denominator, so a rollout that skips the hard tests is scored on what it did not skip. Both are reachable in principle, so we audited: across the five training logs that retain generated code there are **zero** occurrences of a printed ```N passed''`, of `pytest.skip`, or of `sys.exit`, and `critic/rewards/mean` oscillates in 0.46–0.73 without the step change a discovered exploit produces. **The audit’s own limit is that full rollout text was not retained**, so it covers only the fragments those logs carry. Neither seam can reach the reported pass rates, which use a stricter path. Full detail and the fixes we would apply before further training are in §G.13.

**Environment dependence and provenance.** (15) *Serving-stack version dependence.* Every measurement is against vLLM 0.27.1 and verl 0.9.0, so parser behaviour, template defaults and the semantics of `strict: true` are properties of those versions. We pin versions, parser and template hashes, and model revisions for exactly this reason, and we have not replicated on a second serving stack (e.g. SGLang), so we cannot say which findings are vLLM-specific. (16) *Repository-revision dependence.* Chat templates ship inside `tokenizer_config.json` and can be changed by the publisher without a model change; the Qwen2.5-Coder false positive is precisely an artefact of an inherited template. (17) *Instrumentation errata, disclosed rather than absorbed.* Seven ReAct arms recorded `max_tokens=2048` while running at 1024 (an asymmetry that *favoured* FC; re-running changed final pass by  $\leq 1$  item); Llama’s initiation rate required manual correction from 97 to 74 after tool-name validation was added; five arms are error-bearing and admissible only for the error-rate census, and one previously reported  $p=0.648$  computed on such an arm is withdrawn. All are enumerated in Appendix A. None is silently absorbed into a reported number.

## What would change our conclusions (restated)

We name these so the claims are falsifiable rather than merely hedged. *A broken-FC versus repaired-FC RL comparison*, everything else fixed. The repaired arm requires registering a parser that accepts the model’s actual output format in verl’s own registry — a code change, not a different backend. It must be run at **7B**, because the 1.5B checkpoint emits no well-formed call for a repaired parser to accept, and therefore under parameter-efficient tuning, because 7B full-parameter RL does not fit our single card: two arms differing only in that registration. **If repaired-FC recovers multi-turn learning, our sufficiency claim in §4.9 is wrong**; if it does not, the “bottleneck is not singular” reading strengthens from one arm to two, the second being single-variable. This is the experiment most likely to overturn something we have written, which is why we name it precisely. *A second family’s scale ladder*, with a working test executor: if the undercount does not grow with checkpoint size in another family, §4.3’s scale claim is a Qwen-plus-hermes fact rather than a scaling phenomenon. *The same ladder on a non-code task.* The five Qwen2.5-Coder checkpoints run against  $\tau$ -bench **retail** under the documented configuration — models and interface fixed, only the domain varied. If the emitted column still rises with size while the parsed column stays at zero, the scale result is a property of the interface; if it does not, it is a property of code-shaped payloads. This needs a rescue arm at the small sizes for the reason Granite did: a checkpoint that simply never attempts a call produces an uninformative zero, not a censored one. *The same five-layer measurement on further standard benchmarks.* Done for BFCL v4 and for  $\tau$ -bench **retail**

(§4.1), so the measurement-validity concern is *not* specific to our harness, and it now covers one interactive suite with a simulated user and stateful environment. What the  $\tau$ -bench arm does *not* establish is an outcome effect: 7 versus 10 tasks solved on three discordant pairs is far from significant, and its user simulator is a local Llama-3.1-8B rather than **gpt-4o**, so its absolute scores are not comparable to the published leaderboard. Suites whose tool schemas are much larger, or whose scoring depends on argument-level matching, remain untested.